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Method of reducing left atrial pressure

Abstract

A blood flow pathway is formed between a left atrium and a coronary sinus for relieving elevated left atrial pressure. The treatment method includes providing first and second delivery catheters, wherein a medical device is carried by the second delivery catheter. A side outlet opening of the first delivery catheter is positioned within the coronary sinus and oriented toward a left atrium. An anchor on the first delivery catheter contacts an inner surface of the coronary sinus for stabilization. A distal portion of the second delivery catheter is advanced through the side outlet opening. A puncture component is then passed through a target wall portion for creating the blood flow pathway. The medical device is positioned in the opening.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/744,127, filed May 13, 2022, which is a continuation of International Patent Application No. PCT/US2020/059304, filed Nov. 6, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/935,214, filed Nov. 14, 2019, the disclosures of each of which are hereby incorporated by reference.

BACKGROUND

Field

(1) The present disclosure generally relates to the field of transcatheter delivery of medical implant devices and/or therapies.

Description of Related Art

(2) Transcatheter delivery of medical implant devices and/or therapies to a target vessel, channel, chamber and/or organ can be performed to treat various ailments. Delivery of medical implant devices and/or therapies to the heart can be performed to address various heart conditions, including elevated pressure in the left atrium.

SUMMARY

(3) Described herein are systems and methods for providing minimally invasive transcatheter delivery of medical implant devices and/or therapies, including delivery of a medical implant device to the left atrial wall for alleviating elevated left atrial pressure.

(4) In some implementations, a medical implant delivery system can comprise a puncture needle, a medical implant delivery catheter, a medical implant device, and an outer delivery catheter. The puncture needle can comprise an elongate portion and a puncture component associated with a distal end of the elongate portion, the puncture component being configured to pierce and form an opening in a target tissue wall. The medical implant delivery catheter can comprise a puncture needle lumen, the puncture needle being configured to be advanced through the puncture needle lumen, the puncture component being configured to be extended through a puncture needle outlet opening associated with a distal portion of the medical implant delivery catheter. The medical implant device can be positioned on the medical implant delivery catheter. The outer delivery catheter can comprise a medical implant delivery lumen, the medical implant delivery catheter being configured to be advanced through the medical implant delivery lumen, the distal portion of the medical implant delivery catheter being configured to be received within the medical implant delivery lumen and extended through a side outlet opening associated with a distal portion of the outer delivery catheter.

(5) In some embodiments, the medical implant delivery system can comprise a puncture needle sheath configured to be passed through the puncture needle lumen, wherein the puncture needle is configured to be passed through a lumen of the puncture needle sheath. A radially expandable member can be associated with a distal portion of the puncture needle sheath, wherein the radially expandable member is configured to be positioned within the opening formed in the target tissue wall and to enlarge the opening when the radially expandable member is in an expanded configuration. In some embodiments, the medical implant delivery system can comprise a medical implant delivery guide wire, wherein the medical implant delivery guide wire is configured to be slidably advanced through the lumen of the puncture needle sheath and to be exchangeable with the puncture needle.

(6) In some embodiments, a distal portion of the medical implant delivery catheter is configured to be inserted into the opening formed in the target tissue wall, the distal portion comprising a predetermined diameter configured to enlarge the opening.

(7) In some embodiments, the medical implant delivery system can comprise an outer delivery catheter guide wire, and wherein the outer delivery catheter comprises a guide wire lumen configured to receive the outer delivery catheter guide wire. In some embodiments, the outer delivery catheter guide wire lumen extends along an off-center longitudinal axis of the outer delivery catheter.

(8) In some embodiments, a distal portion of the medical implant delivery catheter comprises a lateral cross section having a shape configured to interlock with a shape of the side outlet opening on the distal portion of the outer delivery catheter to orient the medical implant delivery catheter as the medical implant delivery catheter extends through the side outlet opening, wherein the lateral cross section extends along a lateral dimension of the distal portion of the outer delivery catheter.

(9) In some embodiments, the medical implant delivery catheter comprises a first tapered distal portion comprising a first pre-formed curvature and the outer delivery catheter comprises a second

tapered distal portion comprising a second pre-formed curvature, wherein the first curvature and the second curvature comprise the same orientation. In some embodiments, the first pre-formed curvature comprises a radius of curvature smaller than that of the second pre-formed curvature. In some embodiments, the second pre-formed curvature comprises a shape configured to conform to at least a portion of a curvature along a length of a coronary sinus. In some embodiments, the side outlet opening is on an inner edge of the second pre-formed curvature of the outer delivery catheter. (10) In some embodiments, the puncture needle comprises a third pre-formed curvature on a distal portion of the puncture needle. In some embodiments, the third pre-formed curvature comprises a radius of curvature smaller than that of the first pre-formed curvature of the medical implant delivery catheter.

(11) In some embodiments, the medical implant delivery system can comprise an expandable anchor, wherein the expandable anchor and the side outlet opening are on opposing portions of the outer delivery catheter, and wherein the expandable anchor is configured to assume an expanded configuration to position the outer delivery catheter against the target tissue wall. In some embodiments, the outer delivery catheter comprises a second tapered distal portion comprising a second pre-formed curvature and wherein the anchor is on an outer edge of the second pre-formed curvature.

(12) In some implementations, a method for delivering a medical implant device can comprise advancing a guide wire into a vessel, and advancing a medical implant delivery system along the guide wire into the vessel. The medical implant delivery system can comprise an outer delivery catheter comprising a medical implant delivery lumen, a medical implant delivery catheter comprising a medical implant device positioned thereon and a puncture needle lumen extending therethrough, the medical implant delivery catheter slidably extending through the medical implant delivery lumen, and a puncture needle slidably extending through the puncture needle lumen, and a puncture component of the puncture needle extending through a distal outlet opening on a distal end of the medical implant delivery catheter. The method can comprise advancing the medical implant delivery catheter through a side outlet opening on a distal portion of the outer delivery catheter and piercing tissue at a target tissue site using the puncturing component extending through the distal outlet opening to form an opening in the tissue. A distal portion of the medical implant delivery catheter can be extended through the opening formed in the tissue; and the medical implant device positioned on the distal portion of the medical implant delivery catheter can be deployed into the opening formed in the tissue.

(13) In some embodiments, the method can comprise inserting a radially expandable member positioned on a distal portion of a puncture needle sheath in the opening formed in the tissue, and expanding the radially expandable member to enlarge the opening, wherein the puncture needle sheath slidably extends through the puncture needle lumen, and the puncture needle slidably extends through a lumen of the puncture needle sheath.

(14) In some embodiments, the method can comprise exchanging the puncture needle for a medical implant delivery guide wire and inserting the medical implant delivery guide wire through the lumen of the puncture needle sheath.

(15) In some embodiments, the method can comprise inserting a distal portion of the medical implant delivery catheter into the opening formed at the target tissue site to enlarge the opening.

(16) In some embodiments, the method can comprise orienting a distal portion of the medical implant delivery catheter within the medical implant delivery lumen and extending the distal portion through the side outlet opening, wherein orienting comprises rotating the distal portion of the medical implant delivery catheter to match a shape of a lateral cross section of the distal portion with a shape of the side outlet opening, and wherein the lateral cross section extends along a lateral dimension of the distal portion of the outer delivery catheter.

(17) In some embodiments, the method can comprise expanding an expandable anchor on the distal portion of the outer delivery catheter to contact the expandable anchor with a wall of the vessel.

(18) In some embodiments, positioning the guide wire into a vessel comprises positioning the guide wire transfemorally into a coronary sinus through a right atrium via an ostium of the coronary sinus. In some embodiments, piercing tissue at the target tissue site comprises piercing tissue of a left atrial wall to form the opening in the left atrial wall, and wherein deploying the medical implant device comprises deploying the medical implant device into the opening formed in the left atrial wall.

(19) In some embodiments, a medical implant delivery system can comprise a puncture needle, a medical implant delivery catheter, a medical implant device, an elongate housing, and an outer sheath. The puncture needle can comprise an elongate portion and a puncture component at a distal end of the elongate portion, the puncture component being configured to pierce tissue to form an opening in the tissue. The medical implant delivery catheter can comprise a puncture needle lumen and the puncture needle being configured to slidably extend through the puncture needle lumen, and the puncture component being configured to extend through a puncture needle outlet opening at a distal end of the medical implant delivery catheter. The medical implant device can be carried on the medical implant delivery catheter. The elongate housing can be adjacent to and be slidable relative to the medical implant delivery catheter, wherein a distal end of the elongate housing is distal of the distal end of the medical implant delivery catheter when the medical implant delivery catheter is in a retracted configuration. The outer sheath can be around a portion of the medical implant delivery catheter and the elongate housing to maintain the medical implant delivery catheter adjacent to the elongate housing, wherein the outer sheath is slidable relative to the medical implant delivery catheter and extends over the distal end of the medical implant delivery catheter when the medical implant delivery catheter is in the retracted configuration.

(20) In some embodiments, the elongate housing comprises a recess along at least a portion of a length of the elongate housing, a portion of the medical implant delivery catheter being received in the recess. In some embodiments, the elongate housing comprises a guide wire lumen configured to receive an elongate housing guide wire, and the distal end of the elongate housing comprises a guide wire outlet opening configured to allow extension therethrough of the elongate housing guide wire.

(21) In some embodiments, the medical implant delivery catheter comprises a first tapered distal portion comprising a first pre-formed curvature and the elongate housing comprises a second tapered distal portion comprising a second pre-formed curvature, wherein the first pre-formed curvature and the second pre-formed curvature comprise the same orientation. In some embodiments, the first pre-formed curvature comprises a radius of curvature smaller than that of the second pre-formed curvature. In some embodiments, the second pre-formed curvature comprises a shape configured to conform to a curvature along a length of a coronary sinus.

(22) In some embodiments, the medical implant delivery system can comprise an expandable anchor on the distal portion of the elongate housing opposite a portion of the elongate housing adjacent to the medical implant delivery catheter, the expandable anchor being configured to assume an expanded configuration to position the medical implant delivery catheter against a target tissue site. In some embodiments, the elongate housing comprises a second tapered distal portion comprising a second pre-formed curvature and the expandable anchor is on an outer edge of the second pre-formed curvature.

(23) In some embodiments, the medical implant delivery system can comprise a puncture needle sheath slidably extending through the puncture needle lumen, and the puncture needle sheath slidably extending through the puncture needle. A radially expandable member can be on a distal portion of the puncture needle sheath, the radially expandable member being configured to enlarge the opening formed in the tissue.

(24) In some embodiments, the puncture needle is configured to be exchangeable for a medical implant delivery guide wire.

(25) In some implementations, a method for delivering a medical implant device can comprise

positioning a guide wire into a vessel, and advancing a medical implant delivery system along the guide wire into the vessel. The medical implant delivery system can comprise an elongate housing, a medical implant delivery catheter adjacent to the elongate housing, the medical implant delivery catheter carrying a medical implant device and comprising a puncture needle lumen, a puncture needle slidably extending through the puncture needle lumen, and a puncture component of the puncture needle extending through a distal outlet opening on a distal end of the medical implant delivery catheter, and an outer sheath around the elongate housing and the medical implant delivery catheter. The method can comprise proximally sliding the outer sheath to release a distal portion of the medical implant delivery catheter, distally sliding the medical implant delivery catheter relative to the elongate housing and piercing tissue at a target tissue site to form an opening in the tissue using the puncture component extending through the distal outlet opening of the medical implant delivery catheter, extending the distal portion of the medical implant delivery catheter through the opening in the tissue, and deploying the medical implant device carried on the medical implant delivery catheter into the opening.

(26) In some embodiments, the method can comprise positioning a radially expandable member on a distal portion of a puncture needle sheath in the opening formed in the tissue, and expanding the radially expandable member to enlarge the opening, wherein the puncture needle sheath slidably extends through the puncture needle lumen, and the puncture needle slidably extends through a lumen of the puncture needle sheath. In some embodiments, the method can comprise exchanging the puncture needle for a medical implant delivery guide wire and inserting the medical implant delivery guide wire through the lumen of the puncture needle sheath.

(27) In some embodiments, the method can comprise inserting the distal portion of the medical implant delivery catheter into the opening formed at the target tissue site to enlarge the opening.

(28) In some embodiments, the method can comprise expanding an expandable anchor on a distal portion of the elongate housing to contact the expandable anchor with a wall of the vessel.

(29) In some embodiments, positioning the guide wire into the vessel comprises positioning the guide wire transfemorally into a coronary sinus through a right atrium via an ostium of the coronary sinus. In some embodiments, piercing tissue at the target tissue site comprises piercing tissue of a left atrial wall, and wherein deploying the medical implant device comprises deploying the medical implant device into the opening formed in the left atrial wall.

(30) For purposes of summarizing the disclosure, certain aspects, advantages and novel features have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, the disclosed embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments are depicted in the accompanying drawings for illustrative purposes and should in no way be interpreted as limiting the scope of the inventions. In addition, various features of different disclosed embodiments can be combined to form additional embodiments, which are part of this disclosure. Throughout the drawings, reference numbers may be reused to indicate correspondence between reference elements. However, it should be understood that the use of similar reference numbers in connection with multiple drawings does not necessarily imply similarity between respective embodiments associated therewith. Furthermore, it should be understood that the features of the respective drawings are not necessarily drawn to scale, and the illustrated sizes thereof are presented for the purpose of illustration of inventive aspects thereof. Generally, certain of the illustrated features may be relatively smaller than as illustrated in some

embodiments or configurations.

(2) FIG. 1 is a cross-sectional view of a human heart.

(3) FIG. 2 is another cross-sectional view of the human heart.

(4) FIG. 3 is a side view of an example of a medical implant delivery system in accordance with one or more embodiments.

(5) FIG. 4 is another side view of the medical implant delivery system of FIG. 3, in accordance with one or more embodiments.

(6) FIG. 5 is a side view of an outer delivery catheter of the medical implant delivery system of FIG. 3, in accordance with one or more embodiments.

(7) FIG. 6 is a side view of a medical implant delivery catheter of the medical implant delivery system of FIG. 3, in accordance with one or more embodiments.

(8) FIG. 7 shows a proximal handle of the medical implant delivery system of FIG. 3, in accordance with one or more embodiments.

(9) FIGS. 8A and 8B are longitudinal cross-sectional views of the medical implant delivery system of FIG. 3 in a first configuration and a second configuration, respectively, in accordance with one or more embodiments.

(10) FIG. 9 shows a lateral cross-sectional view of the outer delivery catheter of FIG. 5, in accordance with one or more embodiments.

(11) FIG. 10A is a top-down view, and FIG. 10B is a longitudinal cross-sectional view, of an outer delivery catheter of an example of a medical implant delivery system in accordance with one or more embodiments.

(12) FIGS. 10C and 10D provide top-down and side views, respectively, of a medical implant delivery catheter of the medical implant delivery system of FIGS. 10A and 10B, in accordance with one or more embodiments.

(13) FIG. 11 shows insertion of a guide wire into the coronary sinus, in accordance with one or more embodiments.

(14) FIG. 12 shows advancement of the medical implant delivery system of FIG. 3 into the coronary sinus, in accordance with one or more embodiments.

(15) FIG. 13 shows use of the puncture needle of the medical implant delivery system of FIG. 3 to puncture tissue at the target tissue site on the left atrial wall, in accordance with one or more embodiments.

(16) FIG. 14 shows the puncture needle of the medical implant delivery system of FIG. 3 inserted into the left atrium, in accordance with one or more embodiments.

(17) FIG. 15 shows deployment of a shunt device onto the left atrial wall using the medical implant delivery system of FIG. 3, in accordance with one or more embodiments.

(18) FIG. 16 shows withdrawal of the medical implant delivery system of FIG. 3 from the coronary sinus after deployment of the medical implant device onto the left atrial wall, in accordance with one or more embodiments.

(19) FIG. 17 is a process flow diagram of an example of a process to deploy the medical implant delivery system of FIG. 3 to deliver a medical implant device, in accordance with one or more embodiments.

(20) FIG. 18 is a side view of another example of a medical implant delivery system, in accordance with one or more embodiments.

(21) FIG. 19 is a side view of the medical implant delivery system of FIG. 18 comprising the puncture needle in an extended configuration, in accordance with one or more embodiments.

(22) FIG. 20 is a side view of the medical implant delivery system of FIG. 18 with the outer sheath translated proximally, in accordance with one or more embodiments.

(23) FIG. 21 shows advancement of the medical implant delivery system of FIG. 18 into the coronary sinus, in accordance with one or more embodiments.

(24) FIG. 22 shows proximal translation of the outer sheath and distal translation of the medical

implant delivery catheter of the medical implant delivery system of FIG. **18**, while the medical implant delivery system is positioned in the coronary sinus, in accordance with one or more embodiments.

(25) FIG. **23** shows the puncture needle of the medical implant delivery system of FIG. **18** inserted into the left atrium, in accordance with one or more embodiments.

(26) FIG. **24** shows deployment of a shunt device onto the left atrial wall using the medical implant delivery system of FIG. **18**, in accordance with one or more embodiments.

(27) FIG. **25** is a process flow diagram of an example of a process to deploy the medical implant delivery system of FIG. **18** to deliver a medical implant device, in accordance with one or more embodiments.

DETAILED DESCRIPTION

(28) The headings provided herein are for convenience only and do not necessarily affect the scope or meaning of the claimed invention.

(29) The present disclosure relates to systems and methods for providing minimally invasive transcatheter delivery of medical implant devices and/or therapies to a target tissue site on a vessel, channel, chamber and/or organ, including delivery of a shunt device to a target tissue wall, such as a left atrial wall.

(30) Although certain preferred embodiments and examples are disclosed below, inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses and to modifications and equivalents thereof. Thus, the scope of the claims that may arise herefrom is not limited by any of the particular embodiments described below. For example, in any method or process disclosed herein, the acts or operations of the method or process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various operations may be described as multiple discrete operations in turn, in a manner that may be helpful in understanding certain embodiments; however, the order of description should not be construed to imply that these operations are order dependent. Additionally, the structures, systems, and/or devices described herein may be embodied as integrated components or as separate components. For purposes of comparing various embodiments, certain aspects and advantages of these embodiments are described. Not necessarily all such aspects or advantages are achieved by any particular embodiment. Thus, for example, various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may also be taught or suggested herein.

(31) Certain standard anatomical terms of location are used herein to refer to the anatomy of animals, and namely humans, with respect to the preferred embodiments. Although certain spatially relative terms, such as “outer,” “inner,” “upper,” “lower,” “below,” “above,” “vertical,” “horizontal,” “top,” “bottom,” and similar terms, are used herein to describe a spatial relationship of one device/element or anatomical structure to another device/element or anatomical structure, it is understood that these terms are used herein for ease of description to describe the positional relationship between element(s)/structures(s), as illustrated in the drawings. It should be understood that spatially relative terms are intended to encompass different orientations of the element(s)/structures(s), in use or operation, in addition to the orientations depicted in the drawings. For example, an element/structure described as “above” another element/structure may represent a position that is below or beside such other element/structure with respect to alternate orientations of the subject patient or element/structure, and vice-versa.

(32) Various features of a heart **1** are described with reference to FIG. **1** to assist in understanding the present disclosure. The heart **1** includes four chambers, namely the left atrium **2**, the left ventricle **3**, the right ventricle **4**, and the right atrium **5**. A wall of muscle, referred to as the septum **10**, separates the left atrium **2** and right atrium **5**, and the left ventricle **3** and right ventricle **4**. Blood flow through the heart **1** is at least partially controlled by four valves, the mitral valve **6**,

aortic valve **7**, tricuspid valve **8**, and pulmonary valve **9**. The mitral valve **6** separates the left atrium **2** and the left ventricle **3** and controls blood flow therebetween. The aortic valve **7** separates and controls blood flow between the left ventricle **3** and the aorta **12**. The tricuspid valve **8** separates the right atrium **5** and the right ventricle **4** and controls blood flow therebetween. The pulmonary valve **9** separates the right ventricle **4** and the pulmonary artery **11**, controlling blood flow therebetween.

(33) In a healthy heart, the heart valves can properly open and close in response to a pressure gradient present during various stages of the cardiac cycle (e.g., relaxation and contraction) to at least partially control the flow of blood to a respective region of the heart and/or to blood vessels. Deoxygenated blood arriving from the rest of the body generally flows into the right side of the heart for transport to the lungs, and oxygenated blood from the lungs generally flows into the left side of the heart for transport to the rest of the body. During ventricular diastole, deoxygenated blood arrive in the right atrium **5** from the inferior vena cava **15** and superior vena cava **16** to flow into the right ventricle **4**, and oxygenated blood arrive in the left atrium **2** from the pulmonary veins to flow into the left ventricle **3**. During ventricular systole, deoxygenated blood from the right ventricle **4** can flow into the pulmonary artery **11** for transport to the lungs (e.g., via the left **14** and right **13** pulmonary arteries), and oxygenated blood can flow from the left ventricle **3** to the aorta **12** for transport to the rest of the body.

(34) A number of conditions can contribute to a higher than normal pressure in the left atrium. Dysfunction of the mitral valve can contribute to elevated left atrial pressure. Conditions such as mitral valve regurgitation and/or stenosis may result in difficulty in pumping blood from the left atrium to the left ventricle, contributing to elevated pressure in the left atrium. Valve stenosis can cause a valve to become narrowed or obstructed. Mitral valve stenosis can restrict blood flow from the left atrium to the left ventricle. Valve regurgitation occurs when a valve does not close properly. For example, regurgitation can occur due to improper coaptation of the valve leaflets. Mitral valve regurgitation can result in blood flow leakage back into the left atrium **2** from the left ventricle **3** when the left ventricle **3** contracts. Restricted flow of blood from the left atrium **2** into the left ventricle **3**, and blood flow leakage from the left ventricle **3** back into the left atrium **2** can both contribute to elevated atrial pressure. Dysfunction in the left ventricle **3** can also contribute to elevated left atrial pressure. Elevated left atrial pressure may lead to left atrial enlargement, producing symptoms such as shortness of breath during exertion, fatigue, chest pain, fainting, abnormal heartbeat, and swelling of the legs and feet.

(35) FIG. **2** is another view of the heart **1** and shows the coronary sinus **18** around the left atrium **2**. To alleviate elevated left atrial pressure, a conduit can be provided to allow blood flow from the left atrium **2** into a portion of the heart with lower pressure, such as the coronary sinus **18**. A conduit can be formed on the wall of the left atrium **2** adjacent to the coronary sinus **18** to allow blood flow from the left atrium **2** into the coronary sinus **18**. The coronary sinus **18** receives blood from coronary veins and empties into the right atrium **5**. Blood diverted into the coronary sinus **18** from the left atrium **2** can then be delivered into the right atrium **5**. A shunt device, including an expandable shunt device, can be positioned at a location on the left atrial wall, such as a location which is accessible from the coronary sinus **18**, to form a blood flow pathway from the left atrium **2** into the coronary sinus **18**. Access into the coronary sinus **18** can comprise navigating into the right atrium **5** and entering through the coronary sinus ostium **17**.

(36) Traditional minimally invasive transcatheter deployment of a medical implant device and/or therapy to a target tissue site, including transcatheter delivery of a shunt device to the left atrial wall, can include separately advancing a puncture needle delivery catheter and a medical implant delivery catheter. After an opening is formed at the target tissue site using the puncture needle, the puncture needle delivery catheter can be exchanged for the medical implant delivery catheter to deliver the medical implant device and/or therapy. The medical implant device and/or therapy can be delivered to a target tissue site in the heart for altering blood flow in the heart to treat various

abnormal heart conditions. For example, a shunt device can be delivered to the left atrial wall to provide a blood flow pathway between the left atrium and the coronary sinus for relieving elevated left atrial pressure. Delivery of the shunt device to the left atrial wall can require formation of an opening in the left atrial wall. Typically, a puncture needle delivery catheter and a shunt device delivery catheter are separately introduced for forming the opening at the target tissue site and for positioning the shunt device into the opening, respectively. The puncture needle delivery catheter can be retracted and exchanged for the shunt device delivery catheter. Requiring exchange of the shunt device delivery catheter with the puncture needle delivery catheter can result in undesired trauma to patients, increased procedural complexity and/or increased procedural time.

(37) The present disclosure provides systems and methods relating to minimally invasive transcatheter delivery of medical implant devices and/or therapies to a target tissue site. In some embodiments, described herein are medical implant delivery systems configured to enable simultaneously positioning to or proximate to a target tissue site both a puncture needle for piercing the tissue at the target tissue site and a medical implant device and/or therapy for deployment to the target tissue site. The puncture needle and the medical implant device and/or therapy can be delivered together to or proximate to the target tissue site without having to exchange a puncture needle delivery catheter for a delivery catheter to deliver the medical implant device and/or therapy. Delivering the puncture needle and medical implant device and/or therapy together to the target tissue site, without having to separately advance a puncture needle delivery catheter and a delivery catheter for the medical implant device and/or therapy, can facilitate reduced trauma to the patient, provide a simplified procedure, and/or improved procedure time. In some embodiments, use of one or more systems described herein can facilitate up to 50% reduction in procedure time.

(38) In some embodiments, provided herein are medical implant delivery systems for delivering a puncture needle and a shunt device, such as an expandable shunt device, to a target tissue wall, including the left atrial wall. In some embodiments, the medical implant delivery systems can be configured to be positioned into the coronary sinus to access a portion of the left atrial wall from within the coronary sinus. For example, a medical implant delivery system can be advanced into the coronary sinus from the right atrium via the coronary sinus ostium. The right atrium can be accessed via the superior vena cava (SVC) or via the inferior vena cava (IVC). A transjugular or trans-subclavian approach can be used to access the right atrium via the superior vena cava. Alternatively, a transfemoral approach can be used to position the medical implant delivery system into the inferior vena cava, and from the inferior vena cava into the right atrium.

(39) In some embodiments, a medical implant delivery system as described herein can comprise an outer delivery catheter and a medical implant delivery catheter slidably extending through a medical implant delivery lumen of the outer delivery catheter. The medical implant delivery catheter can comprise a puncture needle lumen configured to slidably receive a puncture needle. The puncture needle can extend through an opening on a distal end of the medical implant delivery catheter. A medical implant device can be positioned on the medical implant delivery catheter, such as on a distal portion of the medical implant delivery catheter. For example, a shunt device, such as an expandable shunt device, can be circumferentially positioned around the distal portion of the medical implant delivery catheter.

(40) The outer delivery catheter can be advanced along a guide wire into the coronary sinus. The distal end of the medical implant delivery catheter can be positioned at or proximate to an outlet opening of the outer delivery catheter. In some embodiments, the medical implant delivery catheter can be pre-loaded into the outer delivery catheter. In some embodiments, the medical implant delivery catheter may not be pre-loaded and instead can be advanced from a proximal portion of the outer delivery catheter to the outlet opening after the outer delivery catheter is positioned at a desired location within the coronary sinus. A distal portion of the medical implant delivery catheter can be extended through the outlet opening to enable contact between the puncture needle and the target tissue site so as to form an opening at the target tissue site. After the puncture needle pierces

the tissue, the medical implant delivery catheter can be further extended such that the distal portion of the medical implant delivery catheter extends through the opening formed at the target tissue site so as to deploy the shunt device into the opening, thereby enabling deployment of the shunt device without exchanging a delivery catheter for deploying the shunt device with a delivery catheter for delivering the puncture needle.

(41) In some embodiments, a medical implant delivery system described herein can comprise an elongate housing, a medical implant delivery catheter extending alongside the elongate housing, and an outer sheath around the elongate housing and the medical implant delivery catheter. A medical implant device, including a shunt device, such as an expandable shunt device, can be positioned on the medical implant delivery catheter. For example, the shunt device can be circumferentially around a distal portion of the medical implant delivery catheter. The medical implant delivery catheter can comprise a puncture needle lumen and a puncture needle can be slidably extended through the puncture needle lumen. The puncture needle can be configured to extend through an opening on a distal end of the medical implant delivery catheter.

(42) The elongate housing and the medical implant delivery catheter can be advanced along a guide wire into the coronary sinus. While being advanced to a desired location within the coronary sinus, the outer sheath can be positioned over the distal end of the medical implant delivery catheter to maintain the medical implant delivery catheter alongside the elongate housing. The outer sheath can be subsequently translated proximally relative to the medical implant delivery catheter to release the medical implant delivery catheter. The medical implant delivery catheter can be extended distally to enable contact between the puncture needle and the target tissue site so as to form an opening at the target tissue site. After the puncture needle pierces the tissue, the medical implant delivery catheter can be further extended such that the distal portion of the medical implant delivery catheter extends through the opening formed at the target tissue site so as to deploy the shunt device into the opening, thereby enabling deployment of the shunt device without having to exchange a delivery catheter for deploying the shunt device with a delivery catheter for delivering the puncture needle.

(43) The term “associated with” is used herein according to its broad and ordinary meaning. For example, where a first feature, element, component, device, or member is described as being “associated with” a second feature, element, component, device, or member, such description should be understood as indicating that the first feature, element, component, device, or member is physically coupled, attached, or connected to, integrated with, embedded at least partially within, or otherwise physically related to the second feature, element, component, device, or member, whether directly or indirectly.

(44) Reference herein to “catheters” and/or “delivery catheters” can refer or apply generally to any type of elongate tubular delivery device comprising an inner lumen configured to slidably receive instrumentation, such as for positioning within an atrium or coronary sinus, including for example delivery sheaths and/or cannulas.

(45) FIG. 3 is a side view of an example of a medical implant delivery system **100** comprising an outer delivery catheter **200** and a medical implant delivery catheter **300** slidably extending through a medical implant delivery lumen of the outer delivery catheter **200**. The medical implant delivery catheter **300** can comprise a puncture needle lumen configured to slidably receive a puncture needle **400**. In some embodiments, the medical implant delivery system **100** can comprise a puncture needle sheath (not shown) extending through the puncture needle lumen and the puncture needle **400** can extend through the puncture needle sheath. The puncture needle **400** can be delivered to a target tissue site so as to form an opening on the tissue. A medical implant device **350** (not shown) can be positioned on the medical implant delivery catheter **300** for deployment to the target tissue site. The medical implant device **350** can be positioned on the medical implant delivery catheter **300** such that a delivery catheter for delivering the puncture needle **400** does not need to be exchanged for a delivery catheter for delivering the medical implant device **350**.

(46) In FIG. 3, a distal portion **302** of the medical implant delivery catheter **300** is shown as extending through a side outlet opening **214** on a distal portion **202** of the outer delivery catheter **200**, and the puncture needle **400** is shown as having a portion extending beyond a distal end **310** of the medical implant delivery catheter **300**. The distal portion **202** of the outer delivery catheter **200** is shown in FIG. 3. As shown in the figure, the distal portion **202** can comprise a pre-formed curvature **204**. The pre-formed curvature **204** can be configured to facilitate positioning of the outer delivery catheter **200** into a vessel, channel, chamber and/or organ so as to access the target tissue site. The target tissue site may be a part of the same vessel, channel, chamber and/or organ, or a different vessel, channel, chamber and/or organ. The target tissue site may be on a wall of a vessel, channel, chamber and/or organ. For example, the outer delivery catheter **200** can be positioned into a coronary sinus to access a portion of a left atrial wall accessible from within the coronary sinus. The side outlet opening **214** can be on an inner edge **206** of the pre-formed curvature **204**. In some embodiments, the distal portion **202** can comprise one or more radiopaque markers thereon to facilitate visualization of the outer delivery catheter **200** for positioning the outer delivery catheter **200** at the desired location within the vessel, channel, chamber and/or organ.

(47) A guide wire **50** can be slidably extended through a guide wire lumen of the outer delivery catheter **200**. The guide wire **50** is shown in FIG. 3 as extending beyond a distal end **210** of the outer delivery catheter **200**. The guide wire **50** can be positioned into the vessel, channel, chamber and/or organ such that the outer delivery catheter **200** can be subsequently advanced along the guide wire **50** into the vessel, channel, chamber and/or organ.

(48) In some embodiments, the outer delivery catheter **200** can comprise an expandable anchor **250**. In some embodiments, the expandable anchor **250** can comprise an expandable balloon anchor. The expandable anchor **250** can comprise any number of configurations which enables controlled triggering and/or activation such that the expandable anchor **250** can assume an expanded state from a collapsed state, and vice versa. The expandable anchor **250** is shown in FIG. 3 in a collapsed state. The expandable anchor **250** can be on an outer edge **208** of the pre-formed curvature **204**. For example, the expandable anchor **250** can be on a portion of the outer delivery catheter **200** opposite that of the side outlet opening **214**. In some embodiments, the expandable anchor **250** can be inserted to or proximate to the target tissue site in a collapsed configuration. The expandable anchor **250** can be subsequently triggered and/or actuated to assume an expanded configuration so as to contact tissue near the target tissue site to facilitate stable positioning of the outer delivery catheter **200**. Reliable positioning of the outer delivery catheter **200** relative to the target tissue site can facilitate puncture of the tissue at the target tissue site. In some embodiments, the expandable anchor **250** can press against tissue near the target tissue site, including tissue at an opposing location relative to the target tissue site. Expansion of the expandable anchor **250** can push the delivery catheter **200** toward tissue adjacent to the target tissue, for example pressing the delivery catheter **200** against tissue adjacent to the target tissue. The inner edge **206** portion comprising the side outlet opening **214** of the outer delivery catheter **200** can be positioned against tissue adjacent to the target tissue site such that the puncture needle **400** can be reliably positioned at the target tissue site as the puncture needle **400** is advanced through the side outlet opening **214**. In some embodiments, the expandable anchor **250** can be expanded such that the inner edge **206** of the delivery catheter **200** can be pressed against a portion of the left atrial wall adjacent to the target tissue site. The side outlet opening **214** can thereby be reliably positioned at and/or around the target tissue site on the left atrial wall. The expandable anchor **250** can resume a collapsed state for retraction from the target tissue site.

(49) In some embodiments, an outer delivery catheter may not have an expandable anchor. In some embodiments, an outer delivery catheter comprising an expandable anchor can be selected for a patient with a vessel, channel, chamber and/or organ having a wider diameter at or proximate to the target tissue site.

(50) FIG. 4 is another side view of the medical implant delivery system **100**. The distal portion **302**

of the medical implant delivery catheter **300** is shown extending beyond the side outlet opening **214**. The expandable anchor **250** is shown in FIG. **4** in an expanded state. As described in further detail herein, the medical implant delivery catheter **300** can comprise a pre-formed curvature **304** configured to facilitate delivery of the puncture needle **400** to the target tissue site and/or deployment of the medical implant device **350** positioned on the medical implant delivery catheter **300**.

(51) A puncture needle outlet opening **312** can be associated with the distal portion **302**, including the distal end **310**, of the medical implant delivery catheter **300**. The puncture needle **400** can comprise an elongate portion **402**. A distal portion **404** of the elongate portion **402** is shown extending through the puncture needle outlet opening **312**. A puncture component **408** associated with a distal end **406** of the elongate portion **402** can be used to pierce the tissue at the target tissue site to form the opening in the tissue. In some embodiments, the distal portion **404** of the puncture needle **400** can comprise a pre-formed curvature **410**. In some embodiments, the pre-formed curvature **410** of the puncture needle **400** can facilitate piercing of tissue at the target tissue site. A target tissue site can be to a side of the outer delivery catheter **200** (e.g., at or proximate to the side outlet opening **214**). As described herein, the medical implant delivery catheter **300** can be advanced through the side outlet opening **214** to access the target tissue site. The puncture needle **400** can be advanced through the puncture needle outlet opening **312** of the medical implant delivery catheter **300** to puncture the tissue. The curvature in the pre-formed curvature **410** can facilitate access of the target tissue site to the side of the outer delivery catheter **200** as the puncture needle **400** is advanced through the puncture needle outlet opening **312**. Improved access of the puncture needle **400** to the target tissue site to a side of the outer delivery catheter **200** can provide effective puncturing of the tissue.

(52) In some embodiments, the pre-formed curvature **410** can have the same or similar orientation as the pre-formed curvature **204** of the outer delivery catheter **200**. In some embodiments, the pre-formed curvature **410** can have the same or similar orientation as the pre-formed curvature **304** of the medical implant delivery catheter **300**. In some embodiments, the pre-formed curvature **410** can comprise a radius of curvature smaller than that of the pre-formed curvature **304**.

(53) In some embodiments, the medical implant delivery catheter **300** can be pre-loaded into the medical implant delivery lumen of the outer delivery catheter **200**. The pre-loaded medical implant delivery catheter **300** can remain within the outer delivery catheter **200** until the outer delivery catheter **200** is positioned at or proximate to the target tissue site. For example, the distal end **310** of the medical implant delivery catheter **300** can remain within the medical implant delivery lumen and may not extend beyond the side outlet opening **214** until the outer delivery catheter **200** is positioned within the desired vessel, channel, chamber and/or organ. The distal end **310** of the medical implant delivery catheter **300** may not extend beyond the side outlet opening **214** while the medical implant delivery catheter **300** is in a retracted configuration. In the retracted state, the distal end **310** may be positioned proximal of, at, or distal of the side outlet opening **214**. As described herein, the distal end **310** of the medical implant delivery catheter **300** can be pre-loaded and positioned adjacent to and proximal of the side outlet opening **214**. In some embodiments, the distal end **310** of the medical implant delivery catheter **300** can be pre-loaded and positioned adjacent to and distal of the side outlet opening **214**. In some embodiments, the medical implant delivery catheter **300** may not be pre-loaded and can be advanced from a proximal portion of the outer delivery catheter **200** to the side outlet opening **214** after the outer delivery catheter **200** has been positioned in the desired vessel, channel, chamber and/or organ.

(54) In some embodiments, the puncture needle **400** can be pre-loaded into the puncture needle lumen of the medical implant delivery catheter **300**. In some embodiments, the puncture needle **400** can remain within the medical implant delivery catheter **300** until the outer delivery catheter **200** is positioned at or proximate to the target tissue site. For example, the puncture component **408** may not extend beyond the puncture needle outlet opening **312** until the outer delivery catheter **200** is

positioned within the vessel, channel, chamber and/or organ. The puncture component **408** can remain within the medical implant delivery catheter **300** while the puncture needle is in a retracted configuration. In some embodiments, a portion of the puncture needle **400** can be extended through the puncture needle outlet opening **312** while the puncture needle **400** is in the retracted configuration, but that portion of the puncture needle **400** can remain within the outer delivery catheter **200**. For example, the puncture component **408** can extend through the puncture needle outlet opening **312** while the puncture needle **400** is in the retracted configuration. The puncture component **408** extending beyond the puncture needle outlet opening **312** can reside within the medical implant delivery lumen of the outer delivery catheter **200** until the outer delivery catheter **200** is positioned into the desired vessel, channel, chamber and/or organ, including after the side outlet opening **214** is positioned at a desired position within the vessel, channel, chamber and/or organ. In some embodiments, the puncture needle **400** may not be pre-loaded and can be advanced from a proximal portion of the medical implant delivery catheter **300** to the puncture needle outlet opening **312** after the outer delivery catheter **200** has been positioned in the desired vessel, channel, chamber and/or organ.

(55) In some embodiments, the distal portion **404** of the puncture needle **400** can comprise a shape memory material. As described herein, the distal portion **404** can remain within the medical implant delivery catheter **300** until the outer delivery catheter **200** is positioned at or proximate to the target tissue site, such as the side outlet opening **214** of the outer delivery catheter **200**. The distal portion **404** can assume a configuration having a reduced curvature while received within the medical implant delivery catheter. In some embodiments, the distal portion **404** can assume a linear or substantially linear configuration while received within the medical implant delivery catheter. After extension of the distal portion **404** through the puncture needle outlet opening **312**, the pre-formed curvature **410** on the distal portion **404** can assume the curved configuration. The distal portion **404** can again assume the reduced curvature configuration, including the linear or substantially linear configuration, after retraction back into the medical implant delivery catheter **300**.

(56) In some embodiments, the medical implant delivery system **100** can comprise a radially expandable member (not shown) configured to be inserted into and dilate the opening formed in the tissue by the puncture needle **400**. Enlargement of the opening can facilitate positioning of the medical implant device **350** (not shown) into the opening. The radially expandable member can be associated with a distal portion of a puncture needle sheath (not shown). For example, the puncture needle sheath can extend through the puncture needle lumen of the medical implant delivery catheter **300** and the puncture needle **400** can extend through the puncture needle sheath. For example, at least a portion of the puncture needle **400** can extend within the puncture needle sheath and at least a portion of the puncture needle sheath can extend within the medical implant delivery catheter **300**. In some embodiments, the distal portion **302** of the medical implant delivery catheter **300** can be configured to provide desired dilation of the opening. For example, the distal portion **302** can be inserted into the opening to enlarge the opening. The distal portion **302** can comprise a size and/or shape to facilitate dilation of the opening, for example, having a predetermined diameter configured to provide the desired enlargement. In some embodiments, both the radially expandable member and the distal portion **302** can be used to dilate the opening. For example, the radially expandable member can be used to provide an initial dilation of the opening to facilitate insertion of the distal portion **302** into the opening to provide any desired remaining enlargement of the opening.

(57) In some embodiments, the puncture needle **400** can be exchanged for a medical implant guide wire configured to be inserted through the opening formed at the target tissue site to guide deployment of the medical implant device **350**. For example, the puncture needle **400** can be retracted, and the medical implant guide wire can be advanced through the puncture needle lumen or a puncture needle sheath extending through the puncture needle lumen. The medical implant delivery catheter **300** can be advanced along the medical implant guide wire such that the distal

portion **302** of the medical implant delivery catheter **300** can be inserted through the opening and the medical implant device **350** can be positioned into the opening. In some embodiments, the puncture needle **400** can be further advanced into the opening after formation of the opening to guide subsequent advancement of the medical implant delivery catheter **300**, rather than guide wire. The medical implant delivery catheter **300** can be advanced along the puncture needle **400** such that the medical implant device **350** can be positioned into the opening.

(58) FIG. 5 is a side view of the distal portion **202** of the outer delivery catheter **200**. A guide wire outlet opening **212** can be associated with the distal end **210** of the outer delivery catheter **200**. For example, the guide wire outlet opening **212** can be at the distal end **210**. As described herein, the guide wire **50** can extend through the guide wire lumen of the outer delivery catheter **200**. The guide wire lumen can extend along a length of the outer delivery catheter **200** such that the outer delivery catheter **200** can be passed along the guide wire **50** to advance the outer delivery catheter **200** to a desired location within the vessel, channel, chamber and/or organ. In some embodiments, the guide wire lumen can extend along an entire or substantially an entire length of the outer delivery catheter **200**. For example, the guide wire **50** can extend through an entire or substantially an entire length of the outer delivery catheter **200** and exit through the guide wire outlet opening **212** at the distal end **210**.

(59) The side outlet opening **214** can be on the distal portion **202** of the outer delivery catheter **200**. As described herein, the distal portion **202** of the outer delivery catheter **200** can comprise a pre-formed curvature **204**. The side outlet opening **214** can be on the pre-formed curvature **204**, for example on an inner edge **206** of the pre-formed curvature **204**. For example, after the outer delivery catheter **200** is advanced to a desired location within the desired vessel, channel, chamber and/or organ, the inner edge **206** can be oriented towards the target tissue site, orienting the side outlet opening **214** towards the target tissue, such that the medical implant delivery catheter **300** can be extended through the side outlet opening **214** towards the target tissue site. The puncture needle **400** can then be deployed to contact tissue at the target tissue site. The outer edge **208** of the pre-formed curvature **204** can be oriented away from the target tissue site, such as oriented toward an opposing location relative to the target tissue site.

(60) In some embodiments, the distal portion **202** can comprise a taper. A size of the distal portion **202** can taper toward the distal end **210**. For example, a diameter of the outer delivery catheter **200** can narrow toward the distal end **210**. The taper in the distal portion **202** and/or the pre-formed curvature **204** can be configured to facilitate positioning of the outer delivery catheter **200** into a desired vessel, channel, chamber and/or organ. In some embodiments, a radius of curvature and/or a length of the pre-formed curvature **204** can be selected based at least in part on a shape and/or dimension of the desired vessel, channel, chamber and/or organ. In some embodiments, the medical implant delivery system **100** can be configured to be positioned within the coronary sinus. As described in further detail herein, the medical implant delivery system **100** can be positioned into the coronary sinus via the coronary sinus ostium. The outer delivery catheter **200** can be positioned within the coronary sinus to access a site on the left atrial wall accessible from within the coronary sinus. The tapering in the distal portion **202** can facilitate insertion into narrower portions of the coronary sinus. The length of the pre-formed curvature **204** can be selected based at least in part on a distance of the target tissue site from the coronary sinus ostium. The radius of curvature of the pre-formed curvature **204** can be selected based at least in part on a shape of the coronary sinus, including a degree of curvature of the coronary sinus, in which the outer delivery catheter **200** is positioned. The radius of curvature of the pre-formed curvature **204** can be selected such that the pre-formed curvature **204** can conform or substantially conform to the curvature of a curve of the coronary sinus adjacent into which the outer delivery catheter **200** is positioned. The pre-formed curvature **204** can be positioned such that the inner edge **206** of the pre-formed curvature **204** is oriented towards a portion of the left atrial wall and the outer edge **208** of the curvature is oriented towards a portion of the wall of the coronary sinus, such as an opposing portion of the wall of the

coronary sinus. For example, the pre-formed curvature **204** can follow or substantially follow the curve. In some embodiments, the distal portion **202** of the outer delivery catheter **200** can comprise a shape memory material. For example, the pre-formed curvature **204** can assume the curved configuration after the outer delivery catheter **200** is advanced into the coronary sinus.

(61) FIG. **6** is a side view of the distal portion **302** of the medical implant delivery catheter **300** and the puncture needle **400** extending through the puncture needle outlet opening **312** associated with the distal end **310** of the medical implant delivery catheter **300**. The medical implant delivery catheter **300** can comprise the pre-formed curvature **304** on the distal portion **302**. The pre-formed curvature **304** can have a same or similar orientation as the pre-formed curvature **204** of the outer delivery catheter **200**. For example, when the medical implant delivery catheter **300** is positioned in the coronary sinus, an inner edge **306** of the pre-formed curvature **304** can be oriented towards the left atrial wall and an outer edge **308** of the pre-formed curvature **304** can be oriented towards an opposing portion of a wall of the coronary sinus. In some embodiments, the pre-formed curvature **304** can have a smaller radius of curvature than the pre-formed curvature **204**.

(62) The medical implant device **350** can be carried by the medical implant delivery catheter **300**. Referring to FIG. **6**, in some embodiments, the medical implant device **350** can be positioned on the pre-formed curvature **304**. The medical implant device **350** can be circumferentially positioned around the distal portion **302**, including on the pre-formed curvature **304** of the pre-formed curvature. In some embodiments, the medical implant device **350** can comprise a shunt device, including an expandable shunt device. For example, the expandable shunt device can be configured for delivery to the left atrial wall for addressing elevated left atrial pressure. The expandable shunt device can be positioned around a portion of the distal portion **302**, such as around a portion of the pre-formed curvature **304** of the distal portion **302**.

(63) The expandable shunt device can have a number of configurations. In some embodiments, the expandable shunt device can comprise an expandable tubular shunt device. Examples of suitable expandable shunt devices for the medical implant delivery system **100** are provided in U.S. patent application Ser. No. 15/335,891, entitled "Systems for Deploying an Expandable Cardiac Shunt," which is incorporated herein in its entirety.

(64) The pre-formed curvature **304** can facilitate accessing the target tissue site while the medical implant delivery system **100** is positioned in the vessel, channel, chamber and/or organ. For example, the pre-formed curvature **304** can facilitate proper positioning of the puncture needle **400** at the target tissue site as the medical implant delivery catheter **300** is extended through the side outlet opening **214** of the outer delivery catheter **200** (e.g., at or proximate to the side outlet opening **214**). The curvature in the distal portion **302** of the implant delivery catheter **300** can facilitate access of the target tissue site to the side of the delivery catheter **200** as the distal portion **302** is advanced through the side outlet opening **214**, thereby facilitating proper positioning of the puncture needle **400**. Proper positioning of the puncture needle **400** can enable effective puncturing of the target tissue site.

(65) In some embodiments, the distal portion **302** can comprise a taper. A size of the distal portion **302** can taper toward the distal end **310**. For example, a diameter of the medical implant delivery catheter **300** can narrow toward the distal end **310**. In some embodiments, the taper of the distal portion **302** and/or the pre-formed curvature **304** can facilitate desired deployment of the medical implant device **350** to the target tissue site. As described in further detail herein, the distal portion **302** can be further extended through the opening formed at the target tissue site, such that the medical implant device **350** can be positioned within the opening and deployed. The taper in the distal portion **302** can ease insertion of the distal portion **302** through the opening formed in the tissue. The pre-formed curvature **304** can enable the distal portion **302** to follow a pre-determined trajectory as the distal portion **302** is extended through the side outlet opening **214** of the outer delivery catheter **200** to access the target site to a side of the outer delivery catheter **200** and such

that the distal portion **302** can be inserted into the opening.

(66) In some embodiments, the distal portion **302** of the medical implant delivery catheter **300** can comprise a shape memory material. As described herein, the distal portion **302** can remain within the outer delivery catheter **200** until the outer delivery catheter **200** is positioned at or proximate to the target tissue site. While in a retracted state within the outer delivery catheter, the pre-formed curvature **304** of the distal portion **302** can comprise a configuration having a reduced curvature relative to its relaxed state, such as when the pre-formed curvature **304** is deployed through the side outlet opening **214**. In some embodiments, the distal portion **302** can assume a linear or substantially linear configuration, while received within the outer delivery catheter **200**. After extension of the distal portion **302** through the side outlet opening **214**, the pre-formed curvature **304** on the distal portion **302** can assume the curved configuration. The pre-formed curvature **304** can again assume the reduced curvature configuration, including the linear or substantially linear configuration, after retraction back into the outer delivery catheter **200**. In some embodiments, the outer delivery catheter **200**, medical implant delivery catheter **300**, and/or puncture needle **400** can be flexible such that the outer delivery catheter **200**, medical implant delivery catheter **300**, and/or puncture needle **400** can conform to the shape of anatomical pathways as the medical implant delivery system **100** is advanced through tortuous pathways.

(67) FIG. 7 is a perspective view of an example of a proximal handle **500** of the medical implant delivery system **100**. The proximal handle **500** can be configured to provide control for the deployment of the medical implant delivery system **100**. The proximal handle **500** can comprise control mechanisms for both puncturing tissue at the target tissue site using the puncture needle **400** and deploying the medical implant device **350** positioned on the medical implant delivery catheter **300**. Use of one proximal handle for both tissue puncture and medical implant device deployment can simplify the process for the operator, thereby facilitating a shortened procedure.

(68) An advancer **502** on the proximal handle **500** can be configured to translate proximally and/or distally through a rear bracket **504**. Advancement and/or retraction of the medical implant delivery catheter **300** and outer delivery catheter **200** can be controlled by the advancer **502**. For example, the advancer **502** can be translated distally to advance the medical implant delivery catheter **300** and the outer delivery catheter **200**. The medical implant delivery catheter **300** and outer delivery catheter **200** can extend through the locking nut **506**. The locking nut **506** can be fixed relative to a forward bracket **508** such that the advancer **502**, the medical implant delivery catheter **300** and outer delivery catheter **200** can be fixed relative to the proximal handle **500**. The proximal handle **500** can comprise a pair of flexible arms **510** configured to allow controlled deployment of the medical implant device **350**. For example, the pair of flexible arms **510** can be used to trigger and/or actuate controlled release of the medical implant device **350** from the medical implant delivery catheter **300** into the opening formed at the target tissue site.

(69) FIGS. 8A and 8B show longitudinal cross-sectional views of the medical implant delivery system **100** in a first configuration and a second configuration, respectively. The longitudinal cross-sectional view can comprise a cross-sectional view of the outer delivery catheter **200** along a longitudinal axis of the outer delivery catheter **200**. The distal portion **302** of the medical implant delivery catheter **300** is shown as being received within the medical implant delivery lumen of the outer delivery catheter **200**. The guide wire **50** can extend through the guide wire lumen **220** of the outer delivery catheter **200**. The medical implant device **350** can be positioned on the distal portion **302** of the medical implant delivery catheter **300**. The expandable anchor **250** is shown in FIGS. 8A and 8B in an expanded state. As described herein, the medical implant delivery catheter **300** can be pre-loaded into the outer delivery catheter **200** such that medical implant delivery catheter **300** can be advanced together with the outer delivery catheter **200** into the desired vessel, channel, chamber and/or organ. FIG. 8A shows the medical implant delivery catheter **300** pre-loaded in a first retracted configuration, and FIG. 8B shows the medical implant delivery catheter **300** pre-loaded in a second retracted configuration. As shown in FIG. 8A, in the first retracted configuration, the

distal end **310** of the medical implant delivery catheter **300** is distal of the side outlet opening **214**. For example, prior to deploying the medical implant delivery catheter **300**, such as while advancing the outer delivery catheter **200** along the guide wire **50** to the desired location, the distal end **310** can be distal of the side outlet opening **214**. In some embodiments, the distal end **310** can be distal of and adjacent to (e.g., adjacent to and in contact with) the side outlet opening **214**. The medical implant delivery catheter **300** can be translated proximally relative to the outer delivery catheter **200** to extend the distal end **310** through the side outlet opening **214** for deployment of the medical implant delivery catheter **300**. Referring to FIG. **8B**, the distal end **310** of the medical implant delivery catheter **300** can be positioned proximal of the side outlet opening **214**. In some embodiments, the distal end **310** can be proximal of and adjacent to (e.g., adjacent to and in contact with) the side outlet opening **214**. To deploy the medical implant delivery catheter **300**, the medical implant delivery catheter **300** can be translated distally relative to the outer delivery catheter **200**. The distal end **310** can be adjacent to the side outlet opening **214** to reduce or minimize the distance needed to advance or retract the medical implant delivery catheter **300** for deployment.

(70) In some embodiments, the guide wire lumen of the outer delivery catheter **200** can extend along an off-center longitudinal axis. Referring to FIG. **9**, a lateral cross-sectional view is shown of the outer delivery catheter **200**. The lateral cross section can be taken along a plane which is perpendicular or substantially perpendicular to a longitudinal axis of the outer delivery catheter **200**. The lateral cross section can include a width of the outer delivery catheter **200**. As shown, the guide wire lumen **220** can extend off-set from the central longitudinal axis (C) of the outer delivery catheter **200**. The medical implant delivery catheter **300** (not shown) can thereby be advanced through the outer delivery catheter **200** off-set from the central longitudinal axis (C). For example, the medical implant delivery lumen can be offset from the central longitudinal axis (C). The off-center guide wire lumen **220** can reduce or prevent axial rotation of the medical implant delivery catheter **300** while the medical implant delivery catheter **300** is positioned within the lumen **220**. For example, advancing the medical implant delivery catheter **300** along a path off-set from the central longitudinal axis (C) can facilitate maintaining axial rotation orientation of the medical implant delivery catheter **300** while the medical implant delivery catheter **300** is extended through tortuous anatomical pathways. Maintaining a desired rotational orientation of the medical implant delivery catheter **300** can facilitate a desired exit trajectory of the medical implant delivery catheter **300** as the medical implant delivery catheter **300** is extended through the side outlet opening **214** (not shown). Maintaining a desired exit trajectory of the medical implant delivery catheter **300** can facilitate effective puncturing of the tissue at the target tissue site using the puncture needle **400** (not shown).

(71) FIGS. **10A** through **10D** show another example of a medical implant delivery system **1000**. The medical implant delivery system **1000** can comprise an outer delivery catheter **1100** and a medical implant delivery catheter **1200**. At least a portion of the medical implant delivery catheter **1200** can be slidably received within the outer delivery catheter **1100**. In some embodiments, at least a portion of the medical implant delivery catheter **1200** can be passed through at least a portion of the outer delivery catheter **1100**. FIG. **10A** is a top-down view of the outer delivery catheter **1100**, and FIG. **10B** is a side cross-sectional view of the outer delivery catheter **1100**. FIG. **10C** is a top-down view of the medical implant delivery catheter **1200** and FIG. **10D** is a side view of the medical implant delivery catheter **1200**. A puncture needle (not shown) can be slidably received within the medical implant delivery catheter **1200**. The puncture needle can be configured to be extended through a puncture needle outlet opening **1212** associated with a distal portion **1202**, including a distal end **1210**, of the medical implant delivery catheter **1200**. In some embodiments, at least a portion of the puncture needle can be passed through at least a portion of the medical implant delivery catheter **1200** and extended through the puncture needle outlet opening **1212**.

(72) In some embodiments, the medical implant delivery catheter **1200** can be pre-loaded within the outer delivery catheter **1100**. In some embodiments, the puncture needle can be pre-loaded

within the medical implant delivery catheter **1200**. The medical implant delivery system **1000** may or may not be used in combination with a puncture needle sheath. For example, the puncture needle sheath may extend slidably within the puncture needle lumen of the medical implant delivery catheter **1200** and the puncture needle may extend slidably within a lumen of the puncture needle sheath. The puncture needle may be pre-loaded within the puncture needle sheath and the puncture needle sheath may be pre-loaded within the medical implant delivery catheter **1200**. Alternatively, the medical implant delivery catheter **1200**, puncture needle and/or puncture needle sheath may not be pre-loaded.

(73) A distal portion **1102** of the outer delivery catheter **1100** can comprise a side outlet opening **1114**. Deployment of the medical implant delivery catheter **1200** can include extending the medical implant delivery catheter **1200** through the side outlet opening **1114**. The distal portion **1102** can comprise a pre-formed curvature **1104**. The pre-formed curvature **1104** can be configured to facilitate positioning of the outer delivery catheter **1100** into a vessel, channel, chamber and/or organ so as to access the target tissue site. The side outlet opening **1114** can be on an inner edge **1106** of the pre-formed curvature **1104**. While the outer delivery catheter **1100** is positioned in a desired orientation at the target position, the inner edge **1106** can be oriented toward the target tissue site. An outer edge **1108** of the pre-formed curvature **1104** can be oriented away from the target tissue site, such as toward an opposing position relative to the target tissue site. The distal portion **1202** of the medical implant delivery catheter **1200** can comprise a pre-formed curvature **1204**. In some embodiments, the pre-formed curvature **1204** can have the same orientation as the pre-formed curvature **1104** when the medical implant delivery catheter **1200** is in the deployed configuration, such as when the distal portion **1202** is extended beyond the side outlet opening **1114**. The pre-formed curvature **1204** can be configured to facilitate access to the target tissue site while the outer delivery catheter **1100** is positioned at the target position. In some embodiments, the outer delivery catheter **1100** and/or the medical implant delivery catheter **1200** can be flexible such that the outer delivery catheter **1100** and/or medical implant delivery catheter **1200** can conform to the shape of anatomical pathways as the medical implant delivery system **1000** is advanced through tortuous pathways. In some embodiments, the medical implant delivery catheter **1200** can comprise a reduced curvature configuration while retracted within the outer delivery catheter **1100**. In some embodiments, the medical implant delivery catheter **1200** can comprise a linear or substantially linear configuration while retracted within the outer delivery catheter **1100** and assume the curved configuration after release from the outer delivery catheter **1100**. In some embodiments, the medical implant delivery catheter **1200** and/or the outer delivery catheter **1100** can comprise shape memory material, such as the distal portions **1202**, **1102**.

(74) The side outlet opening **1114** can have a shape to facilitate extension of the medical implant delivery catheter **1200** through the side outlet opening **1114** in a desired orientation. For example, a distal portion **1202** of the medical implant delivery catheter **1200** can be configured to interlock with the side outlet opening **1114** as the distal portion **1202** is extended through the side outlet opening **1114**. In some embodiments, the side outlet opening **1114** can comprise an asymmetry around a central axis. The asymmetry can reduce or prevent rotation of the distal portion **1202** as the distal portion **1202** is extended through the side outlet opening **1114**. The distal portion **1202** can comprise a corresponding shape to interlock with the shape of the side outlet opening **1114**. For example, the distal portion **1202** can comprise a configuration configured to interlock with the asymmetry of the side outlet opening **1114** as at least a portion of the distal portion **1202** is advanced through the side outlet opening **1114**. In some embodiments, a shape of the distal portion **1202** can match, for example being the same or similar as, a shape of the side outlet opening **1114**. Matching between the shapes of the distal portion **1202** and the side outlet opening **1114** can facilitate interlocking between the two.

(75) The side outlet opening **1114** can comprise a proximal portion **1114a** and a distal portion **1114b**. The proximal portion **1114a** and/or the distal portion **1114b** can comprise a shape to

interlock with the distal portion **1202**. In some embodiments, the distal portion **1114b** can have a width narrower than that of the proximal portion **1114a**. In some embodiments, the distal portion **1114b** can comprise an elongated extension. In some embodiments, the distal portion **1114b** can have a “v” shape. In some embodiments, the proximal portion **1114a** can comprise a segment of a circle. In some embodiments, the proximal portion **1114a** can comprise a segment of an oval.

(76) In some embodiments, the distal portion **1202** of the medical implant delivery catheter **1200** can comprise a shape which interlocks with the shape of the side outlet opening **1114**. The distal portion **1202** can comprise a lateral cross section which has a shape configured to interlock with a shape of the side outlet opening **1114**, where the lateral cross section can extend along a lateral dimension of the distal portion of the outer delivery catheter **1100**. In some embodiments, a cross-sectional shape, such as a shape of a lateral cross section, of the distal portion **1202** can have a shape configured to interlock with the shape of the side outlet opening **1114** to orient the medical implant delivery catheter **1200** as the delivery catheter **1200** is extended through the side outlet opening **1114**. The distal portion **1202** can be rotated prior to extending the distal portion **1202** through the side outlet opening **1114** to properly orient the distal portion **1202** relative to the side outlet opening **1114**, for example matching a shape of the lateral cross section of the distal portion **1202** with a shape of the side outlet opening **1114**. In some embodiments, the lateral cross section can be taken along a plane which divides the medical implant delivery catheter **1200** into an upper portion and a lower portion, the upper portion comprising the inner edge **1206** of the pre-formed curvature **1204** and the lower portion comprising the outer edge **1208** of the pre-formed curvature **1204**. The lateral cross section can be taken along a lateral cross-sectional plane which extends along a longitudinal axis of the medical implant delivery catheter **1200** and which is parallel or substantially parallel to a plane containing at least a portion of the side outlet opening **1114**. For example, the lateral cross section can bisect the medical implant delivery catheter **1200** into the upper portion and the lower portion. For example, the distal portion **1202** can comprise a narrowed distal end portion **1214**. The narrowed distal end portion **1214** can be oriented towards and/or align with the distal portion **1114b** of the side outlet opening **1114** so as to guide the orientation of the medical implant delivery catheter **1200** as the medical implant delivery catheter **1200** is extended through the side outlet opening **1114**. For example, interlocking the distal portion **1202** and the side outlet opening **1114** can enable orienting the inner edge **1206** of the pre-formed curvature **1204** proximally and the outer edge **1208** of the pre-formed curvature **1204** distally as the distal portion **1202** is deployed.

(77) The medical implant delivery system **1000** can comprise other features the same as or similar to those of the medical implant delivery system **100**, such as described with reference to FIGS. **3** through **9**. For example, a puncture needle (not shown) can be received in a puncture needle lumen of the medical implant delivery catheter **1200**. The puncture needle can be extended through the puncture needle outlet opening **1212** associated with the distal end **1210** of the medical implant delivery catheter **1200** for puncturing tissue at the target tissue site. A medical implant device **1250** can be positioned on the distal portion **1202** such that a delivery catheter for delivering the puncture needle does not need to be exchanged for a delivery catheter for delivering the medical implant device **1250**. The medical implant device **1250** can be deployed into the tissue opening formed by the puncture needle. The outer delivery catheter **1100** can comprise a guide wire lumen **1120** configured to receive a guide wire. The guide wire can extend through a guide wire outlet opening **1112** associated with a distal end **1110** of the outer delivery catheter such that the outer delivery catheter **1100** can be advanced along the guide wire for positioning into the desired vessel, channel, chamber and/or organ. For example, the medical implant delivery system **1000** can be configured for positioning into the coronary sinus from the right atrium via the coronary sinus ostium such that the puncture needle extending through the medical implant delivery catheter **1200** can be used to form an opening on a portion of the left atrial wall accessible from within the coronary sinus. The medical implant device **1250** can be a shunt device. The shunt device can be deployed into the

opening for treating elevated atrial pressure. The medical implant delivery catheter **1200** can be advanced further into the opening formed in the left atrial wall such that the medical implant device **1250** can be positioned into the opening.

(78) FIGS. **11** through **16** show various steps of an example of a process for deploying the medical implant device **350** using the medical implant delivery system **100** as described herein. Referring to FIG. **11**, the guide wire **50** can be inserted from the right atrium **5**, through the coronary sinus ostium **17** and into the coronary sinus **18**. The right atrium **5** can be accessed via the superior vena cava or via the inferior vena cava. For example, a transjugular or trans-subclavian approach can be used to access the right atrium **5** via the superior vena cava. The medical implant delivery system **100** can be inserted into the subclavian veins or jugular veins, and advanced into the superior vena cava. Alternatively, a transfemoral approach can be used to position the medical implant delivery system **100** into the inferior vena cava.

(79) As shown in FIG. **12**, the medical implant delivery system **100** can be advanced along the guide wire **50** into the coronary sinus **18**. The medical implant delivery system **100** can be advanced along the guide wire **50** until the medical implant delivery system **100** is at a desired location relative to the left atrial wall. For example, the outer delivery catheter **200** can be advanced along the guide wire **50** to position the side outlet opening **214** at or proximate to the target tissue site on the left atrial wall. The pre-formed curvature **204** of the outer delivery catheter **200** can be positioned to follow the curvature of the coronary sinus **18**. For example, the pre-formed curvature **204** can be positioned such that the inner edge **206** of the pre-formed curvature **204** is oriented towards a portion of the left atrial wall and the outer edge **208** of the curvature is oriented towards an opposing portion of a wall of the coronary sinus **18**.

(80) As described herein, in some embodiments, the medical implant delivery system **100** can have the expandable anchor **250** on the outer edge **208** of the curvature **204**. After the outer delivery catheter **200** is positioned at the desired location, the expandable anchor **250** can be triggered and/or actuated to assume an expanded configuration. In the expanded configuration, the expandable anchor **250** can contact a wall of the coronary sinus **18** to push the outer delivery catheter **200** toward the left atrial wall. For example, the expandable anchor **250** can contact the wall of the coronary sinus **18** such that the outer delivery catheter **200** can be positioned against the wall of the left atrium **2**, thereby facilitating stably positioning the medical implant delivery system **100** at the desired location. In some embodiments, the inner edge **206** of the pre-formed curvature **204** can contact the left atrial wall such that the side outlet opening **214** is positioned at and/or surrounding the target tissue site. In some embodiments, the inner edge **206** may not contact the left atrial wall while the expandable anchor **250** is in the expanded state.

(81) As shown in FIG. **12**, the medical implant delivery catheter **300** can be pre-loaded within the medical implant delivery catheter lumen of the outer delivery catheter **200**, and the puncture needle **400** can be pre-loaded within the puncture needle lumen of the medical implant delivery catheter **300**. While advancing the outer delivery catheter **200** to the desired location, the medical implant delivery catheter **300** can be received within the medical implant delivery catheter lumen of the outer delivery catheter **200**. For example, the distal end **310** of the medical implant delivery catheter **300** can be received within the medical implant delivery catheter lumen while the outer delivery catheter **200** is being advanced to the desired location. The puncture needle **400** can be pre-loaded within the puncture needle lumen of the medical implant delivery catheter **300** such that the puncture component **408** is positioned within the puncture needle lumen, or alternatively, such that the puncture component **408** extends through the puncture needle outlet opening **312** associated with the distal end **310** of the medical implant delivery catheter **300**. While being advanced to the target tissue site, the puncture component **408** can extend through the puncture needle outlet opening **312** and remain within the medical implant delivery catheter lumen of the outer delivery catheter **200**.

(82) Referring to FIG. **13**, after the outer delivery catheter **200** is positioned at the desired location,

the medical implant delivery catheter **300** can be extended through the side outlet opening **214**. After the outer delivery catheter **200** is positioned at the desired location, the distal portion **302** of the medical implant delivery catheter **300**, including the distal end **310**, can be extended through the side outlet opening **214**. In some embodiments, the distal portion **302** can be rotated to properly orient the distal portion **302** relative to the side outlet opening **214** prior to extending the distal portion **302** through the side outlet opening.

(83) As described herein, the puncture needle **400** can be pre-loaded within the puncture needle lumen of the medical implant delivery catheter **300**. The puncture needle can be pre-loaded such that the puncture component **408** extends through the puncture needle outlet opening **312** associated with the distal end **310**. The implant delivery catheter **300** can be extended through the side outlet opening **214** such that the puncture component **408** can be used to contact the wall of the left atrium **2** to puncture the tissue. In some embodiments, the puncture needle **400** can be extended further to complete formation of the opening. In some embodiments, the puncture component **408** can be received within the puncture needle lumen of the implant delivery catheter **300** while the outer delivery catheter **200** is inserted to the desired location. The puncture component **408** can be subsequently extended through the puncture needle outlet opening **312** such that the puncture component **408** can be used to contact the target tissue site.

(84) The pre-formed curvature **304** (not shown) of the medical implant delivery catheter **300** can provide a desired trajectory for the distal portion **302** (not shown) of the medical implant delivery catheter **300** as it is advanced through the side outlet opening **214**. For example, the pre-formed curvature **304** can facilitate contact between the puncture needle **400** and the target tissue site on the left atrial wall at a lateral position relative to the medical implant delivery system **100**. The pre-formed curvature **304** can facilitate subsequent insertion of the distal portion **302** into the opening formed at the target tissue site. The pre-formed curvature **410** (not shown) of the puncture needle **400** can facilitate insertion of the distal portion **404** (not shown) of the puncture needle **400** into the tissue at the target tissue site, thereby providing effective puncture of tissue at the target site at a lateral position relative to the medical implant delivery system **100**.

(85) As described herein, the expandable anchor **250** can assume the expanded configuration after the outer delivery catheter **200** is positioned at a desired location within the coronary sinus **18**. In some embodiments, expansion of the expandable anchor **250** can facilitate reliable positioning of the outer delivery catheter **200** against a wall of the coronary sinus. In some embodiments, the outer delivery catheter **200** can be pushed toward the left atrial wall, for example such that the inner edge **206** of the pre-formed curvature **204** can contact the left atrial wall. In some embodiments, positioning of the outer delivery catheter **200** against the left atrial wall can facilitate puncturing of the tissue using the puncture component **408** when the puncture needle **400** is extended through the side outlet opening **214**. In some embodiments, the medical implant delivery system **100** may not include the expandable anchor **250** and/or the inner edge **206** is not positioned in contact with the left atrial wall. For example, the puncture needle **400** can be advanced further through the side outlet opening **214**, such as compared to the extent to which the puncture needle **400** is advanced if the expandable anchor **250** is used, such that the puncture component **408** can contact the left atrial wall and push against the left atrial wall so to pierce the tissue at the target tissue site.

(86) In some embodiments, the medical implant delivery system **100** can comprise a puncture needle sheath. The puncture needle **400** can extend along a puncture needle lumen of the puncture needle sheath. The puncture needle sheath can be received within the medical implant delivery catheter **300**. For example, at least a portion of the puncture needle sheath can be configured to be passed through at least a portion of the puncture needle lumen of the medical implant delivery catheter **300** and at least a portion of the puncture needle **400** can be passed through at least a portion of a lumen of the puncture needle sheath. In some embodiments, the medical implant delivery system **100** can comprise a radially expandable member associated with a distal portion of the puncture needle sheath, the radially expandable member being configured to enlarge the

opening formed in the tissue. The distal portion of the puncture needle sheath can be advanced through the opening such that at least a portion of the radially expandable member is positioned within the opening. The radially expandable member can subsequently be triggered and/or actuated to assume an expanded configuration so as to provide desired enlargement of the opening. The radially expandable member can be deflated and retracted from the opening after desired enlargement is accomplished. In some embodiments, the distal portion **302** of the medical implant delivery catheter **300** can be inserted into the opening to enlarge the opening, for example without using a radially expandable member associated with a puncture needle sheath. In some embodiments, both the radially expandable member and the distal portion **302** of the medical implant delivery catheter **300** can be used to enlarge the opening.

(87) Referring to FIG. **14**, as described herein, in some embodiments, the puncture needle **400** can be extended into the left atrium **2** to serve as a guide wire along which the medical implant delivery catheter **300** can be advanced so as to deploy the medical implant device **350**. Alternatively, the puncture needle **400** can be retracted after the opening in the left atrial wall is formed. The puncture needle **400** can be exchanged for a medical implant guide wire. The medical implant guide wire can be advanced through the puncture needle lumen of the medical implant delivery catheter **300** into the left atrium **2**. In some embodiments, the medical implant guide wire can be advanced through a puncture needle sheath extending through the puncture needle lumen. In some embodiments, the puncture needle **400** can be exchanged for the medical implant guide wire before or after desired enlargement of the opening on the left atrial wall is achieved. For example, the puncture needle **400** can be retracted and the medical implant guide wire can be inserted in its place prior to activating and/or triggering the radially expandable anchor.

(88) FIG. **15** shows deployment of the medical implant device **350**. The medical implant delivery catheter **300**, such as the distal portion **302** of the medical implant delivery catheter **300**, can be advanced through the opening in the left atrial wall such that the medical implant device **350** can be positioned into the opening. The medical implant delivery catheter **300** is shown as being advanced along the puncture needle **400** extended into the left atrium **2**. As described herein, the medical implant device **350** can comprise an expandable shunt device. The expandable shunt device can be positioned into the opening for shunting blood flow from the left atrium **2** into the coronary sinus **18** so as to alleviate elevated left atrial pressure. Examples of processes for deploying an expandable shunt devices into the left atrial wall are provided in U.S. patent application Ser. No. 15/335,891, entitled “Systems for Deploying an Expandable Cardiac Shunt,” which is incorporated herein in its entirety.

(89) FIG. **16** shows withdrawal of the medical implant delivery system **100** from the coronary sinus **18** along the guide wire **50**, after the medical implant device **350** is positioned into the opening in the left atrial wall. As described herein, the medical implant device **350** can comprise an expandable shunt device. Referring to FIG. **16**, the expandable shunt device deployed into the opening formed in the left atrial wall can allow flow of blood from the left atrium **2** into the coronary sinus **18**. The blood flow can then drain into the right atrium **5**. Creating a blood flow pathway from the left atrium **2** into the coronary sinus **18** can thereby alleviate elevated left atrial pressure. After the expandable shunt device is deployed into the left atrial wall, the puncture needle **400** (not shown) and the medical implant delivery catheter **300** (not shown) can be retracted back into the outer delivery catheter **200**. For example, the puncture needle **400** can be retracted back into the puncture needle lumen of the medical implant delivery catheter **300** and the medical implant delivery catheter **300** can be retracted back into the outer delivery catheter **200**. The expandable anchor **250** can assume a collapsed and/or deflated state prior to being withdrawn from the coronary sinus **8**. Alternatively, in the case that a medical implant guide wire is used, the medical implant guide wire can be retracted. Subsequently, the outer delivery catheter **200** can be retracted along the guide wire **50** out of the coronary sinus.

(90) FIG. **17** is a flow diagram of an example of a process **1600** to deploy a medical implant

delivery system for delivering a medical implant device. The medical implant delivery system can comprise one or more features as described herein. The medical implant delivery system can comprise an outer delivery catheter, and a medical implant delivery catheter configured to extend through a lumen of the outer delivery catheter. The medical implant delivery system can comprise a puncture needle extending through a lumen of the medical implant delivery catheter. In block **1602**, the process **1600** can involve advancing the medical implant delivery system along a guide wire into a vessel, channel, chamber and/or organ. For example, the medical implant delivery system can be positioned into the coronary sinus. The guide wire can be advanced into the coronary sinus and the medical implant delivery system can be advanced into the coronary sinus along the already positioned guide wire. The outer delivery catheter can comprise a guide wire lumen and the guide wire can slidably extend through the guide wire lumen.

(91) In block **1604**, the process **1600** can involve advancing the medical implant delivery catheter through a side outlet opening of outer delivery catheter. After the medical implant delivery system is positioned at a desired location within the coronary sinus, the medical implant delivery catheter can be deployed. For example, the medical implant delivery catheter can be extended through a side outlet opening of the outer delivery catheter such that the puncture needle can be used to pierce the tissue at the target tissue site. In block **1606**, the process **1600** can involve extending the puncture component through a distal outlet opening of the medical implant delivery catheter to pierce tissue at the target tissue site to form an opening in the tissue. The target tissue site can be on a portion of the left atrial wall accessible from within the coronary sinus.

(92) In block **1608**, the process **1600** can involve extending a distal portion of medical implant delivery catheter through the opening formed in the tissue. The distal portion can be advanced through the opening to place the medical implant device into the opening. In some embodiments, the distal portion can be configured to enlarge the opening in the tissue. The distal portion can have a predetermined diameter to provide desired enlargement of the opening. In some embodiments, the medical implant delivery system can comprise a puncture needle sheath configured to be slidably passed through the puncture needle lumen of the medical implant delivery catheter, and the puncture needle is configured to be slidably passed through a lumen of the puncture needle sheath. A radially expandable member can be associated with a distal portion of the puncture needle sheath. The radially expandable member can be positioned within the opening formed at in the target tissue site wall to enlarge the opening when the radially expandable member is in an expanded configuration.

(93) In block **1610**, the process **1600** can involve deploying the medical implant device positioned on the medical implant delivery catheter into opening formed in tissue. For example, an expandable shunt device can be deployed into the opening formed on the left atrial wall.

(94) FIGS. **18** through **20** show side views of another example of a medical implant delivery system **1700**. Referring to FIG. **18**, the medical implant delivery system **1700** can comprise an elongate housing **1800**, a medical implant delivery catheter **1900** extending alongside the elongate housing **1800**, and an outer sheath **2200** around the elongate housing **1800** and the medical implant delivery catheter **1900**. A puncture needle sheath **2100** can slidably extend through a puncture needle lumen of the medical implant delivery catheter **1900**. A puncture needle **2000** can slidably extend through the puncture needle sheath **2100**. In some embodiments, the puncture needle **2000** can extend through the puncture needle lumen of the medical implant delivery catheter **1900** without the puncture needle sheath **2100**.

(95) A medical implant device **1950** can be positioned on the medical implant delivery catheter **1900**. The medical implant device **1950** can be positioned on the medical implant delivery catheter **1900** such that a delivery catheter for delivering the puncture needle does not need to be exchanged for a delivery catheter for delivering the medical implant device **1950**. In some embodiments, positioning the medical implant delivery catheter **1900** alongside the elongate housing **1800**, for example rather than extending the medical implant delivery catheter **1900** through an outer delivery

catheter, can provide improved maneuverability in deploying the medical implant delivery catheter **1900**. The medical implant delivery catheter **1900** can be navigated along a tighter curve when deployed, such as to facilitate improved access to a target tissue site. The ability to advance the medical implant delivery catheter **1900** toward and/or into an opening formed in the target tissue without having to extend the medical implant delivery catheter **1900** through a side outlet opening of an outer delivery catheter can facilitate navigation through a tighter curve. Navigation through a tighter curve can thereby improve access to target tissue sites. For example, the ability to navigate through a narrower curvature can enable controlled puncture of target tissue and/or delivery of the medical implant device **1950** for patients with smaller and/or narrower anatomical features at and/or proximate to the target tissue site.

(96) The elongate housing **1800** and the medical implant delivery catheter **1900** can be slidably positioned alongside one another. For example, the medical implant delivery catheter **1900** can be translated proximally and/or distally relative to the elongate housing **1800**. In some embodiments, the elongate housing **1800** and the medical implant delivery catheter **1900** can be adjacent to and in contact with one another. The outer sheath **2200** can be around the elongate housing **1800** and the medical implant delivery catheter **1900** to maintain the position of the elongate housing **1800** and the medical implant delivery catheter **1900** alongside one another.

(97) The medical implant delivery system **1700** can be advanced and/or retracted along the guide wire **50** to and/or from the target location. For example, the elongate housing **1800** can comprise a guide wire lumen extending therethrough configured to receive the guide wire **50**. In some embodiments, the guide wire lumen can extend along an entire length of the elongate housing **1800** and extend through a guide wire outlet opening **1812** associated with a distal end **1810** of the elongate housing **1800**.

(98) The outer sheath **2200** is shown in FIG. **18** as being positioned proximal of the medical implant device **1950**. For example, a distal end **2202** of the outer sheath **2200** is shown as being proximal of the medical implant device **1950**. As described herein, the distal end **2202** can be positioned distal of the medical implant device **1950**, such as while the medical implant delivery system **1700** is advanced to the target location. In some embodiments, the outer sheath **2200** can be maintained over the medical implant device **1950** prior to deployment of the medical implant delivery catheter **1900** and/or medical implant device **1950**. The outer sheath **2200** can be slidable relative to the elongate housing **1800** and the medical implant delivery catheter **1900**. The outer sheath **2200** can be translated proximally and/or distally relative to the medical implant delivery catheter **1900** and the elongate housing **1800**. For example, the outer sheath **2200** can be translated proximally relative to the medical implant delivery catheter **1900** and the elongate housing **1800** to facilitate release a corresponding portion of the medical implant delivery catheter **1900** from being positioned adjacent to the elongate housing **1800**. In some embodiments, after delivery of the medical implant device **1950**, the outer sheath **2200** can be translated distally relative to the medical implant delivery catheter **1900** and the elongate housing **1800** to position the medical implant delivery catheter **1900** adjacent to and alongside the elongate housing **1800** prior to withdrawal of the medical implant delivery system **1700**.

(99) In some embodiments, a distal portion **1802** of the elongate housing **1800** can comprise a taper. A size of the distal portion **1802** can taper toward the distal end **1810**. For example, a diameter of the elongate housing **1800** can narrow toward the distal end **1810**. In some embodiments, the distal portion **1802** of the elongate housing **1800** can comprise a pre-formed curvature **1804**. The pre-formed curvature **1804** and/or the taper in the distal portion **1802** can be configured to facilitate positioning of the elongate housing **1800** into a vessel, channel, chamber and/or organ so as to deliver a medical implant device and/or therapy to a target site accessible from within the vessel, channel, chamber and/or organ. A radius of curvature and/or a length of the pre-formed curvature **1804** can be selected based at least in part on a shape and/or dimension of the vessel, channel, chamber and/or organ, such as the coronary sinus. In some embodiments, the

length of the pre-formed curvature **1804** can be predetermined based at least in part on a distance of the target tissue site from the coronary sinus ostium. In some embodiments, the radius of curvature of the pre-formed curvature **1804** can be predetermined based at least in part on a shape of the coronary sinus, including a degree of curvature of a portion of the coronary sinus in which the elongate housing **1800** is positioned. The radius of curvature of the pre-formed curvature **1804** can be selected such that the pre-formed curvature **1804** can conform or substantially conform to a curvature of the coronary sinus. For example, when positioned in the coronary sinus, the pre-formed curvature **1804** can be positioned such that the inner edge **1806** of the pre-formed curvature **1804** is oriented towards a portion of the left atrial wall and the outer edge **1808** of the pre-formed curvature **1804** is oriented towards an opposing portion of a wall of the coronary sinus. In some embodiments, the elongate housing **1800**, including the distal portion **1802** can comprise a shape memory material. The distal portion **1802** can assume a reduced curvature configuration while being advanced to the target tissue site. The pre-formed curvature **1804** can assume the curved configuration after positioning in the target position. The tapering in the distal portion **1802** can facilitate insertion into narrower portions of the coronary sinus. In some embodiments, the distal portion **1802** can comprise one or more radiopaque markers thereon to facilitate visualization of positioning the elongate housing **1800** at the desired location within the vessel, channel, chamber and/or organ, such as the coronary sinus.

(100) The elongate housing **1800** can comprise a recess **1814** configured to receive at least a portion of the medical implant delivery catheter **1900**. The recess **1814** can extend along at least a portion of a length of the elongate housing **1800**. For example, the recess **1814** can extend from a proximal portion, including a proximal end, to the distal portion **1802** of the elongate housing **1800**. In some embodiments, a distal end **1816** of the recess **1814** can be proximal of the pre-formed curvature **1804**. In some embodiments, the distal end **1816** of the recess **1814** can be on the pre-formed curvature **1804**. The recess **1814** can open along the same side as the inner edge **1806** of the pre-formed curvature **1804**.

(101) In some embodiments, the elongate housing **1800** can comprise an expandable anchor **1850** on the distal portion **1802**. The expandable anchor **1850** is shown in a deflated state in FIG. **18**. In some embodiments, the expandable anchor **1850** can be on the same side as an outer edge **1808** of the pre-formed curvature **1804**. In some embodiments, the expandable anchor **1850** can be on the pre-formed curvature **1804**. For example, the expandable anchor **1850** can be on the outer edge **1808** of the pre-formed curvature **1804**. In some embodiments, the expandable anchor **1850** can be on a portion of the elongate housing **1800** opposite the distal end **1816** of the recess **1814**. In some embodiments, the expandable anchor **1850** can be on a portion of the elongate housing **1800** opposite and proximal of the pre-formed curvature **1804**.

(102) The expandable anchor **1850** can be inserted to or proximate to the target tissue site in a collapsed configuration. The expandable anchor **1850** can be subsequently triggered and/or actuated to assume an expanded configuration so as to contact tissue near the target tissue site to facilitate stable positioning of the elongate housing **1800** relative to the target tissue site. In some embodiments, expansion of the expandable anchor **1850** can facilitate reliable positioning of the elongate housing **1800** against tissue portion at a location opposite that of the target tissue site. In some embodiments, expansion of the expandable anchor **1850** can push the elongate housing **1800** and/or the medical implant delivery catheter **1900** against tissue adjacent to the target tissue site. Reliable positioning of the elongate housing **1800** at the target tissue site can facilitate desired deployment of the puncture needle **2000** to form the opening. In some embodiments, the elongate housing **1800** may not have an expandable anchor **1850**. The puncture needle **2000** can be extended further, such as compared to the extent to which the puncture needle **2000** would be advanced if the expandable anchor **1850** is used, such that the puncture component **2008** can contact and push against the target tissue site to facilitate effective puncture of the tissue. For example, the puncture needle **2000** can be further extended to contact and push against the left atrial wall so as to pierce

the tissue on the left atrial wall.

(103) Referring again to FIG. 18, the distal portion **1902** of the medical implant delivery catheter **1900** can comprise a taper. A size of the distal portion **1902** can taper toward the distal end **1910**, a diameter of the medical implant delivery catheter **1900** narrowing toward the distal end **1910**. In some embodiments, the medical implant delivery catheter **1900** can have a pre-formed curvature **1904** on a distal portion **1902**. The pre-formed curvature **1904** can have a same or similar orientation as the pre-formed curvature **1804** of the elongate housing **1800**. For example, when the medical implant delivery catheter **1900** is positioned in the coronary sinus, an inner edge **1906** of the pre-formed curvature **1904** can be oriented towards the left atrial wall and an outer edge **1908** of the pre-formed curvature **1904** can be oriented towards a wall of the coronary sinus. In some embodiments, the pre-formed curvature **1904** can have a smaller radius of curvature than the pre-formed curvature **1804**.

(104) In some embodiments, the taper of the distal portion **1902** and/or the pre-formed curvature **1904** can facilitate accessing the target tissue site while the medical implant delivery system **1700** is positioned in the vessel, channel, chamber and/or organ. The taper can facilitate effective puncture of the target tissue site and/or deployment of the medical implant device **1950** positioned on the medical implant delivery catheter **1900**. The distal portion **1902** can be further extended through the opening formed at the target tissue site, such that the medical implant device **1950** can be positioned within the opening and deployed. The taper in the distal portion **1902** can ease insertion of the distal portion **1902** through the opening formed in the tissue. The pre-formed curvature **1904** can enable the distal portion **1902** to follow a pre-determined trajectory to facilitate extending the distal portion **1902** through the opening.

(105) The medical implant delivery catheter **1900** can comprise a puncture needle lumen configured to slidably receive the puncture needle **2000**. The puncture needle **2000** can be extended through a puncture needle outlet opening **1912** associated with, the distal portion **1902**, including a distal end **1910**, of the medical implant delivery catheter **1900**. In some embodiments, the pre-formed curvature **1904** can facilitate proper positioning of the puncture needle **2000** at the target tissue site as the medical implant delivery catheter **1900** is advanced relative to the elongate housing **1800**. Proper positioning of the puncture needle **2000** can enable effective puncturing of the target tissue site.

(106) In some embodiments, the medical implant delivery system **1700** can include a puncture needle sheath **2100** configured to allow the puncture needle **2000** to extend therethrough. The puncture needle **2000** can extend through a puncture needle outlet opening **2104** associated with a distal end **2106** of the puncture needle sheath **2100**. The puncture needle sheath **2100** can slidably extend through the puncture needle lumen of the medical implant delivery catheter **1900**. In some embodiments, at least a portion of the puncture needle sheath **2100** can be passed through at least a portion of the puncture needle lumen of the medical implant delivery catheter **1900**. In some embodiments, at least a portion of the puncture needle **2000** can be passed through at least a portion of the lumen of the puncture needle sheath **2100**. In some embodiments, a distal portion **2102** of the puncture needle sheath **2100** can have associated therewith a radially expandable member **2150** configured to enlarge the opening formed at the target tissue site. The radially expandable member **2150** can be triggered and/or actuated after it has been positioned into the opening to assume an expanded configuration so as to enlarge the opening. In some embodiments, the radially expandable member **2150** can comprise an inflatable balloon circumferentially positioned around the distal portion **2102** of the puncture needle sheath **2100**.

(107) In some embodiments, the distal portion **1902** of the medical implant delivery catheter **1900** can be configured to provide desired dilation of the opening. For example, the distal portion **1902** can be inserted into the opening to enlarge the opening. The distal portion **1902** can comprise a size and/or shape to facilitate dilation of the opening, for example, having a predetermined diameter configured to provide the desired enlargement. In some embodiments, the distal portion **1902** can

be used to dilate the opening instead of the radially expandable member **2150**. For example, the puncture needle sheath **2100** and/or the radially expandable member **2150** may not be included. In some embodiments, the puncture needle **2000** can be positioned within the puncture needle lumen of the medical implant delivery catheter **1900** without the puncture needle sheath **2100**. In some embodiments, the distal portion of the puncture needle **2000** can be configured to enlarge the opening. The distal portion can be inserted into the opening to enlarge the opening. For example, the distal portion can have a predetermined diameter to provide desired enlargement of the opening. In some embodiments, one or more of the puncture needle **2000**, the distal portion **1902** and the radially expandable member **2150** can be used to dilate the opening formed in the target tissue.

(108) In some embodiments, the puncture needle **2000** can be exchanged for a medical implant guide wire. For example, the puncture needle **2000** can be retracted after the puncture needle **2000** has been used to pierce the tissue and the opening is formed. The medical implant guide wire can be advanced through the puncture needle lumen in place of the puncture needle **2000**. The medical implant device **1950** can be advanced along the medical implant guide wire to the target tissue site. For example, an expandable shunt device can be advanced along the medical implant guide wire to an opening formed on the left atrial wall. In some embodiments, the medical implant guide wire can be advanced through the puncture needle sheath **2100**. In some embodiments, the medical implant delivery system **1700** may not include the puncture needle sheath **2100**. For example, the puncture needle **2000** and/or the medical implant guide wire can extend through the puncture needle lumen of the medical implant delivery catheter **1900** without the puncture needle sheath **2100**.

(109) The medical implant device **1950** can be positioned on the distal portion **1902** of the medical implant delivery catheter **1900**. In some embodiments, the medical implant device **1950** can comprise a shunt device, including an expandable shunt device. As described in further detail herein, the distal portion **1902** can be further extended through the opening formed at the target tissue site, such that the medical implant device **1950** can be positioned within the opening and deployed. In some embodiments, the medical implant device **1950** can be positioned on the pre-formed curvature **1904**. The medical implant device **1950** can be circumferentially positioned around the distal portion **1902**, including on the pre-formed curvature **1904**.

(110) The expandable shunt device can have a number of configurations. In some embodiments, the expandable shunt device can comprise an expandable tubular shunt device. Examples of suitable expandable shunt devices for the medical implant delivery system **1700** are provided in U.S. patent application Ser. No. 15/335,891, entitled “Systems for Deploying an Expandable Cardiac Shunt,” which is incorporated herein in its entirety.

(111) FIG. **19** is a side view of the medical implant delivery system **1700** where the puncture needle **2000** is further extended through the puncture needle outlet opening **1912**. A portion of the puncture needle **2000** is shown as being extended through puncture needle outlet opening **2104** associated with the distal end **2106** of the puncture needle sheath **2100**. The puncture needle sheath **2100** is shown as being extended through the puncture needle outlet opening **1912** associated with the distal end **1910** of the medical implant delivery catheter **1900**. The expandable anchor **1850** of the elongate housing **1800** is shown in an expanded state.

(112) The puncture needle **2000** can comprise an elongate portion **2002**. A distal portion **2004** of the elongate portion **2002** is shown extending through the puncture needle outlet opening **1912** of the medical implant delivery catheter **1900**. The puncture needle **2000** can have a puncture component **2008** associated with a distal end **2006** of the elongate portion **2002**, the puncture component **2008** being configured for tissue piercing. The puncture needle **2000** can comprise a pre-formed curvature **2010** on the distal portion **2004**. The pre-formed curvature **2010** can have a same or similar orientation as the pre-formed curvature **1904** of the medical implant delivery catheter **1900**. In some embodiments, the pre-formed curvature **2010** can have a smaller radius of

curvature than the pre-formed curvature **1904**. In some embodiments, the pre-formed curvature **2010** can have a same or similar orientation as the pre-formed curvature **1804** of elongate housing **1800**. In some embodiments, the pre-formed curvature **2010** can facilitate effective tissue puncture at the target tissue site while the medical implant delivery system **1700** is positioned in the vessel, channel, chamber and/or organ.

(113) In some embodiments, the distal portion **2004** of the puncture needle **2000** can comprise a shape memory material. The distal portion **2004** can remain within the medical implant delivery catheter **1900** prior to the medical implant delivery system **1700** being positioned at or proximate to the target tissue site and/or after the distal portion **2004** is retracted back into the medical implant delivery catheter **1900**. In some embodiments, the pre-formed curvature **2010** can assume a less curved configuration while retracted within the medical implant delivery catheter **1900** and/or the puncture needle sheath **2100**. In some embodiments, the distal portion **2004** can assume a linear or substantially linear configuration while received within the medical implant delivery catheter **1900** and/or the puncture needle sheath **2100**. The pre-formed curvature **2010** on the distal portion **2004** can assume the curved configuration after extension of the distal portion **2004** through the puncture needle outlet opening **1912** and/or the puncture needle outlet opening **2104** associated with the distal end **2106** of the puncture needle sheath **2100**.

(114) FIG. **20** is another side view of the medical implant delivery system **1700**. The outer sheath **2200** is positioned over the medical implant device **1950**. For example, the medical implant device **1950** is not shown in FIG. **20** as the distal end **2202** of the outer sheath **2200** is distal of the medical implant device **1950**, thereby extending over the medical implant device **1950**. The distal end **1910** of the medical implant delivery catheter **1900** is shown as extending distally of the outer sheath **2200**. The puncture needle **2000** and the puncture needle sheath **2100**, including the radially expandable member **2150**, are shown as extending from puncture needle outlet **1912** associated with the distal end **1910** of the medical implant delivery catheter **1900**. The puncture needle **2000** is shown as extending from the puncture needle outlet opening **2104** associated with the distal end **2106** of the puncture needle sheath **2100**.

(115) In some embodiments, while the medical implant delivery catheter **1900** is in a retracted configuration, the outer sheath **2200** can be positioned over the distal end **1910** of the medical implant delivery catheter **1900**. The distal end **2202** of the outer sheath **2200** can be at or distal of the distal end **1910** of the medical implant delivery catheter **1900**. In some embodiments, the distal portion **1902** can comprise a shape memory material. For example, the pre-formed curvature **1904** can assume a less curved configuration, including a linear or substantially linear configuration, while the outer sheath **2200** is positioned around the pre-formed curvature **1904**. The outer sheath **2200** positioned around the pre-formed curvature **1904** can maintain the pre-formed curvature **1904** in the linear or substantially linear configuration. In some embodiments, the outer sheath **2200**, elongate housing **1800**, and/or the medical implant delivery catheter **1900** can be flexible such that the outer sheath **2200**, elongate housing **1800** and/or medical implant delivery catheter **1900** can conform to the shape of anatomical pathways as the medical implant delivery system **1700** is advanced through tortuous pathways. The outer sheath **2200** can maintain the medical implant delivery catheter **1900** in the recess **1814** of the elongate housing **1800**. The outer sheath **2200** can be configured to be translated proximally relative to the implant delivery catheter **1900** to release the medical implant delivery catheter **1900**. For example, the distal portion **1902** can be displaced from the recess **1814**. The pre-formed curvature **1904** can be allowed to assume the curved configuration, such as after being released from the recess **1814**. The medical implant delivery catheter **1900** can be distally translatable relative to the outer sheath **2200** and/or elongate housing **1800** to position the medical implant device into the opening formed in the tissue.

(116) The expandable anchor **1850** is shown in FIG. **20** an expanded state. The expandable anchor **1850** can remain in a collapsed state until the medical implant delivery system **1700** is positioned at or proximate to the target tissue site. The expandable anchor **1850** can be in the collapsed state

while the medical implant delivery catheter **1900** is in a retracted configuration. In some embodiments, while the medical implant delivery catheter **1900** is in the retracted configuration, the outer sheath **2200** can be positioned over the expandable anchor **1850**. The distal end **2202** of the outer sheath **2200** can be distal of the expandable anchor **1850**. In some embodiments, the outer sheath **2200** can remain proximal of the expandable anchor **1850** while the medical implant delivery catheter **1900** is in the retracted configuration.

(117) In some embodiments, the puncture needle **2000** and/or the puncture needle sheath **2100** can be pre-loaded into the puncture needle lumen of the medical implant delivery catheter **1900**. In some embodiments, a pre-loaded puncture needle **2000** and pre-loaded puncture needle sheath **2100** can remain within the medical implant delivery catheter **1900** until the medical implant delivery system **1700** is positioned at or proximate to the target tissue site. For example, the puncture component **2008** may not extend beyond the puncture needle outlet opening **1912** until the medical implant delivery system **1700** is positioned within the vessel, channel, chamber and/or organ. The puncture component **2008** can remain within the medical implant delivery catheter **1900** while the puncture needle **2000** is in a retracted configuration. In some embodiments, a portion of the puncture needle **2000** can extend through the puncture needle outlet opening **1912** while the puncture needle **2000** is in the retracted configuration. In some embodiments, the puncture component **2008** extending beyond the puncture needle outlet opening **1912** can be proximal of the distal end **2202** of the outer sheath **2200** when in the puncture needle **2000** is in the retracted configuration. For example, the puncture component **2008** can be received within the recess **1814** of the elongate housing **1800** until the medical implant delivery system **1700** is positioned into the desired vessel, channel, chamber and/or organ. In some embodiments, the puncture needle **2000** and/or the puncture needle sheath **2100** can be advanced from a proximal portion of the medical implant delivery catheter **1900** to the puncture needle outlet opening **1912** after the medical implant delivery system **1700** has been positioned in the desired vessel, channel, chamber and/or organ.

(118) FIGS. **21** through **24** show various steps of an example of a process for delivering a medical implant device, such as an expandable shunt device, to a left atrial wall from within the coronary sinus using the medical implant delivery system **1700** as described herein. Referring to FIG. **21**, the medical implant delivery system **1700** can be advanced along the guide wire **50** into the coronary sinus **18**. As described herein, the guide wire **50** can be inserted from the right atrium **5**, through the coronary sinus ostium **17** and into the coronary sinus **18**. In some embodiments, a transjugular or trans-subclavian approach can be used to access the right atrium **5** via the superior vena cava. Alternatively, a transfemoral approach can be used to position the medical implant delivery system **1700** through the inferior vena cava and into the right atrium **5**. The medical implant delivery system **1700** can be positioned at a desired location within the coronary sinus **18**. While the medical implant delivery system **1700** is advanced along the guide wire **50** to the desired location within the coronary sinus **18**, the outer sheath **2200** can be positioned over the distal end **1910** (not shown) of the medical implant delivery catheter **1900** (not shown). The medical implant delivery catheter **1900** can be maintained within the recess **1814** of the elongate housing **1800**. The outer sheath **2200** can be proximal of the expandable anchor **1850**. In some embodiments, the outer sheath **2200** can be positioned over the expandable anchor **1850** while the medical implant delivery system **1700** is positioned to the desired location. In some embodiments, the distal end **2202** of the outer sheath **2200** can be at or distal of the distal end **1816** of the recess **1814** while the medical implant delivery system **1700** is advanced along the guide wire **50**.

(119) In FIG. **22**, after the medical implant delivery system **1700** has been advanced to the desired location, the outer sheath **2200** can be translated proximally relative to the medical implant delivery catheter **1900** and the elongate housing **1800**. The outer sheath **2200** can be slid proximally to release the distal portion **1902** of the medical implant delivery catheter **1900**. The distal end **2202** of the outer sheath **2200** can be translated proximally such that at least a portion of the distal portion **1902** can be released from the recess **1814** of the elongate housing **1800**. For example, the

pre-formed curvature **1904** can assume its curved configuration after the distal portion **1902** is released. The inner edge **1906** of the pre-formed curvature **1904** can be oriented towards the wall of the left atrium **2** and the outer edge **1908** of the pre-formed curvature **1904** can be oriented towards a wall of the coronary sinus **18**. As described herein, in some embodiments, the puncture needle **2000** can be pre-loaded such that the puncture component **2008** extends through the puncture needle outlet opening **1912** of the medical implant delivery catheter **1900**. In some embodiments, the puncture component **2008** can extend beyond the puncture needle outlet opening **2104** of the puncture needle sheath **2100**. For example, the puncture component **2008** can be extended beyond the puncture needle outlet opening **2104** and the puncture needle outlet opening **1912** while remaining proximal of the distal end **2202** of the outer sheath **2200** while the within the medical implant delivery system **1700** is advanced to its target location. Alternatively, the puncture component **2008** can be extended through the puncture needle outlet opening **1912** and/or the puncture needle outlet opening **2104** after the medical implant delivery system **1700** has been positioned at or proximate to the desired location. After the distal portion **1902** is released, the medical implant delivery catheter **1900** can be advanced distally such that the puncture component **2008** can contact the left atrial wall and can be used to pierce the tissue at the target tissue site on the left atrial wall. In some embodiments, the puncture needle sheath **2100** can be advanced, such as relative to the medical implant delivery catheter **1900**, to facilitate contact between the puncture component **2008** and the left atrial wall. The puncture needle **2000** can be further advanced, such as relative to the medical implant delivery catheter **1900** and/or the puncture needle sheath **2100**, to form the opening on the left atrial wall.

(120) In some embodiments, the pre-formed curvature **1904** of the medical implant delivery catheter **1900** can facilitate positioning the puncture needle **2000** at a target tissue site on the left atrial wall and/or subsequent insertion of the distal portion **1902** of the medical implant delivery catheter **1900** into the opening formed at the target tissue site. The pre-formed curvature **1904** can provide a desired trajectory for the medical implant delivery catheter **1900** as it is advanced distally such that the puncture needle **2000** can contact a target tissue site on the left atrial wall at a lateral position relative to the medical implant delivery system **1700** and the distal portion **1902** of the implant delivery catheter **1900** can be inserted into the opening formed at the target tissue site.

(121) In some embodiments, the expandable anchor **1850** can be triggered and/or actuated to assume an expanded state after the medical implant delivery system **1700** is positioned at or proximate to the desired location within the coronary sinus. In the expanded configuration, the expandable anchor **1850** can contact a wall of the coronary sinus **18** to facilitate stably positioning the medical implant delivery system **1700** at the desired location. Stable positioning of the medical implant delivery system **1700** can facilitate tissue puncture at the target tissue site and/or reliable deployment of the medical implant device **1950** into the opening. In some embodiments, expansion of the expandable anchor **1850** can provide stable positioning of the elongate housing **1800** against the coronary sinus **18** so as to facilitate reliable puncture of the target tissue site on the left atrial wall.

(122) Referring to FIG. 23, the puncture needle sheath **2100** can be extended into the opening formed in the left atrial wall such that the radially expandable member **2150** associated with the distal portion **2102** of the puncture needle sheath **2100** can be positioned into the opening to enlarge the opening. Enlargement of the opening can facilitate subsequent positioning of the medical implant device **1950** into the opening. The radially expandable member **2150** can be triggered and/or actuated after it has been positioned into the opening to assume an expanded configuration so as to enlarge the opening. As described herein, in some embodiments, the distal portion **1902** of the medical implant delivery catheter **1900** can be inserted into the opening to provide desired dilation of the opening.

(123) The puncture needle **2000** can be further advanced through the puncture needle outlet opening **2104** associated with the distal end **2106** of the puncture needle sheath **2100**. The puncture

needle **2000** can serve as a guide wire along which the medical implant delivery catheter **1900** can be advanced such that the medical implant device **1950** (not shown) can be positioned into the opening in the tissue. Alternatively, the puncture needle **2000** can be exchanged for a medical implant guide wire. For example, the puncture needle **2000** can be retracted after the opening in the left atrial wall is formed. The medical implant guide wire can be advanced through the puncture needle lumen of the medical implant delivery catheter **1900** into the left atrium **2** and the implant delivery catheter **1900** can be advanced along the medical implant guide wire into the left atrium **2** to deploy the medical implant device **1950**. FIG. **23** shows that the outer sheath **2200** remains over the medical implant device **1950** while the opening is formed in the left atrial wall. In some embodiments, the medical implant device **1950** can be distal of the outer sheath **2200** while the opening is formed in the left atrial wall.

(124) Referring to FIG. **24**, the distal portion **1902** of the medical implant delivery catheter **1900** can be advanced through the opening formed in the left atrial wall such that the medical implant device **1950** can be positioned into the opening. The medical implant delivery catheter **1900** can be advanced along the puncture needle **2000** positioned into the left atrium **2** to position the medical implant device **1950** into the opening. Examples of processes for deploying an expandable shunt devices into the left atrial wall are provided in U.S. patent application Ser. No. 15/335,891, entitled “Systems for Deploying an Expandable Cardiac Shunt,” which is incorporated herein in its entirety.

(125) The expandable shunt device deployed into the opening formed in the left atrial wall can allow flow of blood from the left atrium **2** into the coronary sinus **18**, and then into the right atrium **5**, thereby alleviating elevated left atrial pressure. After the expandable shunt device is deployed into the left atrial wall, the puncture needle **2000** and the medical implant delivery catheter **1900** can be retracted. Subsequently, the medical implant delivery system **1700** can be retracted along the guide wire **50** out of the coronary sinus.

(126) FIG. **25** is a flow diagram of an example of a process **2400** to deploy a medical implant delivery system for delivering a medical implant device. The medical implant delivery system can comprise one or more features as described herein, such as that of the medical implant delivery system **1700** described with reference to FIGS. **18** through **24**.

(127) In block **2402**, the process **2400** can involve advancing the medical implant delivery system along a guide wire into a vessel, channel, chamber and/or organ. For example, the medical implant delivery system can be advanced into the coronary sinus along the guide wire. The medical implant delivery system can be used to access the target tissue site on a target vessel, channel, chamber and/or organ. For example, the medical implant delivery system can be used to access a target tissue site on a portion of the left atrial wall from within the coronary sinus. The medical implant delivery system can comprise an elongate housing, a medical implant delivery catheter extending alongside the elongate housing, and an outer sheath around the elongate housing and the medical implant delivery catheter. A medical implant device can be positioned on the medical implant delivery catheter for deployment onto the target vessel, channel, chamber and/or organ. For example, a shunt device, including an expandable shunt device, can be carried on the medical implant delivery catheter for deployment onto the left atrial wall.

(128) After the medical implant delivery system is positioned at a desired location within the coronary sinus, the puncture needle can be deployed to pierce the tissue at the target tissue site. A puncture needle can slidably extend through a puncture needle lumen of the medical implant delivery catheter. In block **2404**, the process **2400** can involve translating the outer sheath proximally to release a distal portion of medical implant delivery catheter. As described herein, the distal portion of the medical implant delivery catheter can comprise a pre-formed curvature. The distal portion can comprise a shape memory material such that the distal portion can assume the curved configuration after the outer sheath is slid proximally such that the outer sheath is no longer around the distal portion to position the distal portion alongside the elongate housing.

(129) In block **2406**, the process **2400** can involve translating the medical implant delivery catheter

distally relative to the elongate housing. The medical implant delivery catheter can be advanced distally such that the puncture needle can contact the target tissue. In some embodiments, the puncture component of the puncture needle can be extended through a puncture needle outlet opening associated with a distal end of the medical implant delivery catheter to facilitate contact between the puncture component and the target tissue. In block **2408**, the process **2400** can involve piercing tissue at the target tissue site to form an opening in the tissue using the puncture component extending through the distal outlet opening of the medical implant delivery catheter. (130) In block **2410**, the process **2400** can involve extending a distal portion of the medical implant delivery catheter through the opening formed in the tissue. The distal portion of the medical implant delivery catheter can be advanced through the opening such that the medical implant device carried on the medical implant delivery can be positioned in the opening. In some embodiments, the distal portion can be configured to enlarge the opening in the tissue. For example, the distal portion can have a predetermined diameter to provide desired enlargement of the opening. In some embodiments, the medical implant delivery system can comprise a puncture needle sheath configured to slidably extend through the puncture needle lumen of the medical implant delivery catheter, and the puncture needle is configured to be slidably passed through a lumen of the puncture needle sheath. A radially expandable member can be associated with a distal portion of the puncture needle sheath. At least a portion of the radially expandable member can be positioned within the opening formed in the tissue and expanded to enlarge the opening. (131) In block **2412**, the process **2400** can involve deploying the medical implant device positioned on the medical implant delivery catheter can into the opening formed in the tissue. For example, the shunt device can be deployed onto the left atrial wall.

Additional Embodiments

(132) Depending on the embodiment, certain acts, events, or functions of any of the processes or algorithms described herein can be performed in a different sequence, may be added, merged, or left out altogether. Thus, in certain embodiments, not all described acts or events are necessary for the practice of the processes.

(133) Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is intended in its ordinary sense and is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous, are used in their ordinary sense, and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is understood with the context as used in general to convey that an item, term, element, etc. may be either X, Y or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y and at least one of Z to each be present.

(134) It should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, Figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of one or more of the various inventive aspects. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than are expressly recited in that claim. Moreover, any components, features, or steps illustrated and/or described in a particular embodiment herein

can be applied to or used with any other embodiment(s). Further, no component, feature, step, or group of components, features, or steps are necessary or indispensable for each embodiment. Thus, it is intended that the scope of the inventions herein disclosed and claimed below should not be limited by the particular embodiments described above, but should be determined only by a fair reading of the claims that follow.

(135) It should be understood that certain ordinal terms (e.g., “first” or “second”) may be provided for ease of reference and do not necessarily imply physical characteristics or ordering. Therefore, as used herein, an ordinal term (e.g., “first,” “second,” “third,” etc.) used to modify an element, such as a structure, a component, an operation, etc., does not necessarily indicate priority or order of the element with respect to any other element, but rather may generally distinguish the element from another element having a similar or identical name (but for use of the ordinal term). In addition, as used herein, indefinite articles (“a” and “an”) may indicate “one or more” rather than “one.” Further, an operation performed “based on” a condition or event may also be performed based on one or more other conditions or events not explicitly recited.

(136) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(137) The spatially relative terms “outer,” “inner,” “upper,” “lower,” “below,” “above,” “vertical,” “horizontal,” and similar terms, may be used herein for ease of description to describe the relations between one element or component and another element or component as illustrated in the drawings. It be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the drawings. For example, in the case where a device shown in the drawing is turned over, the device positioned “below” or “beneath” another device may be placed “above” another device. Accordingly, the illustrative term “below” may include both the lower and upper positions. The device may also be oriented in the other direction, and thus the spatially relative terms may be interpreted differently depending on the orientations.

(138) Unless otherwise expressly stated, comparative and/or quantitative terms, such as “less,” “more,” “greater,” and the like, are intended to encompass the concepts of equality. For example, “less” can mean not only “less” in the strictest mathematical sense, but also, “less than or equal to.”

Claims

1. A method of forming an opening between a coronary sinus and a left atrium, the method comprising: providing a first delivery catheter and a second delivery catheter pre-loaded within the first delivery catheter, and a medical device carried by a distal portion of the second delivery catheter; advancing the first delivery catheter, the second delivery catheter, and the medical device together from a right atrium of a heart and into the coronary sinus; positioning a side outlet opening of a distal portion of the first delivery catheter within the coronary sinus and orienting the side outlet opening toward the left atrium; expanding an expandable anchor coupled to the first delivery catheter to contact an inner surface of the coronary sinus for stabilizing the first delivery catheter within the coronary sinus; advancing the distal portion of the second delivery catheter outwardly through the side outlet opening toward a target wall portion of the coronary sinus; puncturing the target wall portion with a puncture component associated with a distal end of the second delivery catheter to form the opening between the coronary sinus and left atrium while the medical device carried by the second delivery catheter is positioned in the coronary sinus; and positioning the medical device within the opening between the coronary sinus and the left atrium.

2. The method of claim 1, wherein the opening provides a blood flow pathway between the left atrium and the coronary sinus for relieving elevated left atrial pressure.
 3. The method of claim 1, wherein the medical device has side wall portions comprising a plurality of struts.
 4. The method of claim 1, wherein the first delivery catheter includes a first pre-formed curved section along a distal portion.
 5. The method of claim 4, wherein the distal portion of the second delivery catheter has a second pre-formed curved section, the second pre-formed curved section having a radius of curvature smaller than that of the first pre-formed curved section.
 6. The method of claim 1, wherein the first delivery catheter, the second delivery catheter, and the medical device are advanced together over a guidewire into the coronary sinus.
 7. The method of claim 1, wherein the first delivery catheter includes one or more radiopaque markers for facilitating visualization.
 8. A method of accessing a coronary sinus, the method comprising: advancing at least a portion of a distal portion of a first delivery catheter into a right atrium, through a coronary sinus ostium, and into the coronary sinus from the right atrium; positioning a side outlet opening on the distal portion of the first delivery catheter into the coronary sinus; advancing at least a portion of a distal portion of a second delivery catheter out of the first delivery catheter through the side outlet opening, the distal portion of the second delivery catheter carrying an expandable medical device and having a puncture component associated with a distal end of the second delivery catheter; and deploying the medical device carried on the distal portion of the second delivery catheter to an opening formed in the target tissue; wherein the opening extends from a left atrium into the coronary sinus to provide a blood flow pathway, thereby relieving elevated left atrial pressure.
 9. The method of claim 8, wherein advancing the at least a portion of the distal portion of the first delivery catheter into the coronary sinus comprises orienting an inner edge of a first pre-formed curved distal portion of the first delivery catheter towards a portion of a left atrial wall.
 10. The method of claim 8, wherein deploying the expandable medical device comprises positioning the medical device in the opening formed in the target tissue.
 11. The method of claim 8, wherein the expandable medical device has side wall portions comprising a plurality of struts.
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